

the wage rate in all towns falls with the imposition of the tax in town 1. However, we see that this fall in the wage rate approaches zero as N grows large. Thus, at this point, it may still seem that our partial equilibrium approximation will have given us the correct answers for large N . But this is not so. Consider the effect of the tax on total profits. The partial equilibrium approach told us that workers escaped the tax; all the tax fell as a burden on firms. But letting $\pi(w)$ be the profit function of a representative firm, the change in aggregate profits from the imposition of this tax is¹⁷

$$(N-1)\pi'(\bar{w})w'(0) + \pi'(\bar{w})(w'(0) + 1) = \pi'(\bar{w})\left(-\frac{N-1}{N} + \frac{N-1}{N}\right) = 0.$$

Aggregate profits stay constant! Thus, all of the burden of a small tax falls on laborers, not on the owners of firms. Although the partial equilibrium approximation is correct as far as getting prices and wages about right, it errs by just enough, and in just such a direction, that the conclusions of the tax incidence analysis based on it are completely reversed.¹⁸

17. Recall that the profits of the firm in town 1 are $\pi(w(t) + t)$.

18. We note that the justifications of partial equilibrium analysis in terms of small individual budget shares that we informally described in Sections 3.I and 10.G do not apply here because the "consumption" goods in this example (jobs in different towns) are perfect substitutes and therefore individual budget shares are not guaranteed to be small at all prices.

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EXERCISES

15.B.1^A Consider an Edgeworth box economy in which the two consumers have locally nonsatiated preferences. Let $x_{\ell i}(p)$ be consumer i 's demand for good ℓ at prices $p = (p_1, p_2)$.

(a) Show that $p_1(\sum_i x_{1i}(p) - \bar{\omega}_1) + p_2(\sum_i x_{2i}(p) - \bar{\omega}_2) = 0$ for all prices p .

(b) Argue that if the market for good 1 clears at prices $p^* \gg 0$, then so does the market for good 2; hence, p^* is a Walrasian equilibrium price vector.

15.B.2^A Consider an Edgeworth box economy in which the consumers have the Cobb–Douglas utility functions $u_1(x_{11}, x_{21}) = x_{11}^\alpha x_{21}^{1-\alpha}$ and $u_2(x_{12}, x_{22}) = x_{12}^\beta x_{22}^{1-\beta}$. Consumer i 's endowments are $(\omega_{1i}, \omega_{2i}) \gg 0$, for $i = 1, 2$. Solve for the equilibrium price ratio and allocation. How do these change with a differential change in ω_{11} ?

15.B.3^B Argue (graphically) that in an Edgeworth box economy with locally nonsatiated preferences, a Walrasian equilibrium is Pareto optimal.

15.B.4^C Consider an Edgeworth box economy. An offer curve has the *gross substitute* property if an increase in the price of one commodity decreases the demand for that commodity and increases the demand for the other one.

(a) Represent in an Edgeworth box the shape of an offer curve with the gross substitute property.

(b) Assume that the offer curves of the two consumers have the gross substitute property. Show then that the offer curves can intersect only once (not counting the intersection at the initial endowments).

Let us denote an offer curve as *normal* if an increase in the price of one commodity leads to an increase in the demand for that commodity only if the demands of the two commodities both increase.

(c) Represent in the Edgeworth box the shape of a normal offer curve that does not satisfy the gross substitute property.

(d) Show that there are preferences giving rise to offer curves that are not normal. Show that the demand function for such preferences is not normal (i.e., at some prices some good is inferior).

(e) Show in the Edgeworth box that if the offer curve of one consumer is normal and that of the other satisfies the gross substitute property, then the offer curves can intersect at most once (not counting the intersection at the initial endowments).

(f) Show that two normal offer curves can intersect several times.

15.B.5^A Verify that the offer curves of Example 15.B.2 are as claimed. Solve also for the claimed values of relative prices.

15.B.6^B (D. Blair) Compute the equilibria of the following Edgeworth box economy (there is more than one):

$$\begin{aligned} u_1(x_{11}, x_{21}) &= (x_{11}^{-2} + (12/37)^3 x_{21}^{-2})^{-1/2}, & \omega_1 &= (1, 0), \\ u_2(x_{12}, x_{22}) &= ((12/37)^3 x_{12}^{-2} + x_{22}^{-2})^{-1/2}, & \omega_2 &= (0, 1). \end{aligned}$$

15.B.7^C Show that if both consumers in an Edgeworth box economy have continuous, strongly monotone, and strictly convex preferences, then the Pareto set has no “holes”: precisely, it is a connected set. Show that if, in addition, the preferences of both consumers are homothetic, then the Pareto set lies entirely on one side of the diagonal of the box.

15.B.8^B Suppose that both consumers in an Edgeworth box have continuous and strictly convex preferences that admit a quasilinear utility representation with the first good as numeraire. Show that any two Pareto optimal allocations in the interior of the Edgeworth box then involve the same consumptions of the second good. Connect this with the discussion of Chapter 10.

15.B.9^B Suppose that in a pure exchange economy (i.e., an economy without production), we have two consumers, Alphonse and Betatrix, and two goods, Perrier and Brie. Alphonse and Betatrix have the utility functions:

$$u_\alpha = \text{Min}\{x_{p\alpha}, x_{b\alpha}\} \quad \text{and} \quad u_\beta = \text{Min}\{x_{p\beta}, (x_{b\beta})^{1/2}\}$$

(where $x_{p\alpha}$ is Alphonse's consumption of Perrier, and so on). Alphonse starts with an endowment of 30 units of Perrier (and none of Brie); Betatrix starts with 20 units of Brie (and none of Perrier). Neither can consume negative amounts of a good. If the two consumers behave as price takers, what is the equilibrium?

Suppose instead that Alphonse begins with only 5 units of Perrier while Betatrix's initial endowment remains 20 units of Brie, 0 units of Perrier. What happens now?

15.B.10^C (*The Transfer Paradox*) In a two-consumer, two-commodity pure exchange economy with continuous, strictly convex and strongly monotone preferences, consider the comparative statics of the welfare of consumer 1 with changes in the initial endowments $\omega_1 = (\omega_{11}, \omega_{21})$ and $\omega_2 = (\omega_{12}, \omega_{22})$.

(a) Suppose first that the preferences of the two consumers are quasilinear with respect to the same numeraire. Show that if the endowments of consumer 1 are increased to $\omega'_1 \gg \omega_1$ while ω_2 remains the same, then at equilibrium the utility of consumer 1 *may* decrease. Interpret this observation and relate it to the theory of a quantity-setting monopoly.

(b) Suppose now that the increase in resources of consumer 1 constitute a transfer from consumer 2, that is, $\omega'_1 = \omega_1 + z$ and $\omega'_2 = \omega_2 - z$ with $z \geq 0$. Under the same assumption as in (a), show that the utility of consumer 1 cannot decrease.

(c) Consider again a transfer as in (b), but this time preferences may not be quasilinear. Suppose that the transfer z is small and that similarly the change in the equilibrium (relative) price is restricted to be small. Show that it is possible for the utility of consumer 1 to decrease (this is called the *transfer paradox*). A graphical illustration in the Edgeworth box suffices to make the point. Interpret in terms of the interplay between substitution and wealth effects.

(d) Show that in this Edgeworth box example (but, be warned, not more generally) the transfer paradox can happen only if there is a multiplicity of equilibria. [*Hint*: Argue graphically in the Edgeworth box. Show that if a transfer to consumer 1 leads to a decrease of the utility of consumer 1, then there must be an equilibrium at the no-transfer situation where consumer 1 gets an even lower level of utility.]

15.C.1^B This exercise refers to the one-consumer, one-firm economy discussed in Section 15.C.

(a) Prove that in an economy with one firm, one consumer, and strictly convex preferences and technology, the equilibrium level of production is unique.

(b) Fix the price of output to be 1. Define the excess demand function for labor as

$$z_1(w) = x_1(w, w\bar{L} + \pi(w)) + y_1(w) - \bar{L},$$

where w is the wage rate, $\pi(\cdot)$ is the profit function, and $x_1(\cdot, \cdot)$, $y_1(\cdot)$ are, respectively, the consumer's demand function for leisure and the firm's demand function for labor. Show that the slope of the excess demand function is not necessarily of one sign throughout the range of prices but that it is necessarily negative in a neighborhood of the equilibrium.

(c) Give an example to show that there can be multiple equilibria in a strictly convex economy with one firm and two individuals, each of whom is endowed with labor alone. (Assume that profits are split equally between the two consumers.) Can this happen if the firm operates under constant rather than strictly decreasing returns to scale?

15.C.2^A Consider the one-consumer, one-producer economy discussed in Section 15.C. Compute the equilibrium prices, profits, and consumptions when the production function is $f(z) = z^{1/2}$, the utility function is $u(x_1, x_2) = \ln x_1 + \ln x_2$, and the total endowment of labor is $\bar{L} = 1$.

15.D.1^B In text.

15.D.2^A Show that in the 2×2 production model the production possibility set is convex (assume free disposal).

15.D.3^B Show that the Stolper–Samuelson theorem (Proposition 15.D.1) can be strengthened to assert that the increase in the price of the intensive factor is proportionally larger than the increase in the price of the good (and therefore the well-being of a consumer who owns only the intensive factor must increase).

15.D.4^C Consider a general equilibrium problem with two consumer-workers ($i = 1, 2$), two constant returns firms ($j = 1, 2$) with concave technologies, two factors of production ($\ell = 1, 2$), and two consumption goods ($j = 1, 2$) produced, respectively, by the two firms. Assume that the production of consumption good 1 is relatively more intensive in factor 1. Neither consumer consumes either of the factors. Consumer 1 owns one unit of factor 1 while consumer 2 owns one unit of factor 2.

(a) Set up the equilibrium problem as one of clearing the factor and goods markets (in a closed economy context) under the assumption that prices are taken as given and productions are profit maximizing.

(b) Suppose that consumer 1 has a taste only for the second consumption good and that consumer 2 cares only for the first good. Argue that there is at most one equilibrium.

(c) Suppose now that consumer 1 has a taste only for the first good and that consumer 2 cares only for the second good. Argue that a multiplicity of equilibria is possible.

[Hint: Parts (b) and (c) can be answered by graphical analysis in the Edgeworth box of factors of production.]

15.D.5^B Show that the Rybczynski theorem (Proposition 15.D.2) can be strengthened to assert that the proportional increase in the production of the good that uses the increased factor relatively more intensively is greater than the proportional increase in the endowment of the factor.

15.D.6^C Suppose you are in the 2×2 production model with output prices (p_1, p_2) given (the economy could be a small open economy). The factor intensity condition is satisfied (production of consumption good 1 uses factor 1 more intensively). The total endowment vector is $\bar{z} \in \mathbb{R}^2$.

(a) Set up the equilibrium conditions for factor prices (w_1^*, w_2^*) and outputs (q_1^*, q_2^*) allowing for the possibility of specialization.

(b) Suppose that $\hat{w} = (\hat{w}_1, \hat{w}_2)$ are factor prices with the property that for each of the two goods the unit cost equals the price. Show that the necessary and sufficient condition for the equilibrium determined in (a) to have $(q_1^*, q_2^*) \gg 0$ is that $\bar{z}$ belongs to the set

$$\{(z_1, z_2) \in \mathbb{R}_+^2 : a_{11}(\hat{w})/a_{21}(\hat{w}) > z_1/z_2 > a_{12}(\hat{w})/a_{22}(\hat{w})\},$$

where $a_{\ell j}(\hat{w})$ is the optimal usage (at factor prices $\hat{w}$) of the input ℓ in the production of one unit of good j . This set is called the *diversification cone*.

(c) The unit-dollar isoquant of good j is the set of factor combinations that produce an amount of good j of 1 dollar value. Show that under the factor intensity condition the unit-dollar isoquants of the two goods can intersect at most once. Use the unit-dollar isoquants to construct graphically the diversification cone. [Hint: If they intersect twice then there are two points (one in each isoquant) proportional to each other and such that the slopes of the isoquants at these points are identical.]

(d) When the total factor endowment is not in the diversification cone, the equilibrium is specialized. Can you determine, as a function of total factor endowments, in which good the economy will specialize and what the factor prices will be? Be sure to verify the inequality conditions in (a). To answer this question you can make use of the graphical apparatus developed in (c).

15.D.7^B Suppose there are two output goods and two factors. The production functions for the two outputs are

$$f_1(z_{11}, z_{21}) = 2(z_{11})^{1/2} + (z_{21})^{1/2} \quad \text{and} \quad f_2(z_{12}, z_{22}) = (z_{12})^{1/2} + 2(z_{22})^{1/2}.$$

The international prices for these goods are $p = (1, 1)$. Firms are price takers and maximize profits. The total factor endowments are $\bar{z} = (\bar{z}_1, \bar{z}_2)$. Consumers have no taste for the consumption of factors of production. Derive the equilibrium factor allocation $((z_{11}^*, z_{21}^*), (z_{12}^*, z_{22}^*))$ and the equilibrium factor prices (w_1^*, w_2^*) as a function of $(\bar{z}_1, \bar{z}_2)$. Verify that you get the same result whether you proceed through equations (15.D.1) and (15.D.2) or by solving (15.D.5).

15.D.8^B The setting is as in the 2×2 production model. The production functions for the two outputs are of the Cobb–Douglas type:

$$f_1(z_{11}, z_{21}) = (z_{11})^{2/3}(z_{21})^{1/3} \quad \text{and} \quad f_2(z_{12}, z_{22}) = (z_{12})^{1/3}(z_{22})^{2/3}.$$

The international output price vector is $p = (1, 1)$ and the total factor endowments vector is $\bar{z} = (\bar{z}_1, \bar{z}_2) \gg 0$. Compute the equilibrium factor allocations and factor prices for all possible values of $\bar{z}$. Be careful in specifying the region of total endowment vectors where the economy will specialize in the production of a single good.

15.D.9^C (*The Heckscher–Ohlin Theorem*) Suppose there are two consumption goods, two factors, and two countries A and B. Each country has technologies as in the 2×2 production model. The technologies for the production of each consumption good are the same in the two countries. The technology for the production of the first consumption good is relatively more intensive in factor 1. The endowments of the two factors are $\bar{z}_A \in \mathbb{R}_+^2$ and $\bar{z}_B \in \mathbb{R}_+^2$ for countries A and B, respectively. We assume that country A is relatively better endowed with factor 1, that is, $\bar{z}_{1A}/\bar{z}_{2A} > \bar{z}_{1B}/\bar{z}_{2B}$. Consumers are identical within and between countries. Their preferences are representable by increasing, concave, and homogeneous utility functions that depend only on the amount consumed of the two consumption goods.

Suppose that factors are not mobile and that each country is a price taker with respect to the international prices for consumption goods. Suppose then that at the international prices $p = (p_1, p_2)$ we have that, first, neither of the two countries specializes and, second, the international markets for consumption goods clear. Prove that country A must be exporting good 1, the good whose production is relatively more intensive in the factor that is relatively more abundant in country A.

Equilibrium and Its Basic Welfare Properties

16.A Introduction

With this chapter, we begin our systematic study of equilibrium in economies where agents act as price takers. We consider a world with L commodities in which consumers and firms interact through a market system. In this market system, a price is quoted for every commodity, and economic agents take these prices as independent of their individual actions.

We concentrate in this chapter on a presentation of the basic welfare properties of equilibria. Some more advanced topics in welfare economics are discussed in Chapter 18 and in Part V.

We begin, in Section 16.B, by specifying the formal model of an economy to be studied here and for the rest of Part IV. Its essential ingredients—commodities, consumers, and firms—we have already encountered in Part I. The remainder of Section 16.B introduces the main concepts that will concern us throughout the chapter. We define first the normative notion of a *Pareto optimal allocation*, an allocation with the property that it is impossible to make any consumer better off without making some other consumer worse off. Then, we present two notions of price-taking equilibrium: *Walrasian* (or *competitive*) *equilibrium*, and its generalization, a *price equilibrium with transfers*. The Walrasian equilibrium concept applies to the case of a *private ownership economy*, in which a consumer's wealth is derived from her ownership of endowments and from claims to profit shares of firms. The more general notion of a price equilibrium with transfers allows instead for an arbitrary distribution of wealth among consumers.

The remaining sections of the chapter are devoted to exploring the relationships between these equilibrium concepts and Pareto optimality.

Section 16.C focuses on the statement of the (very weak) conditions implying that every price equilibrium with transfers (and, hence, every Walrasian equilibrium) results in a Pareto optimal allocation. This is the *first fundamental theorem of welfare economics*, a formal expression for competitive market economies of Adam Smith's claimed "invisible hand" property of markets.

In Section 16.D, we study the converse issue. We state conditions (convexity assumptions are the crucial ones) under which every Pareto optimal allocation can be

supported as a price equilibrium with transfers. This result is known as the *second fundamental theorem of welfare economics*. It tells us that if its assumptions are satisfied, then through the use of appropriate lump-sum wealth transfers, a welfare authority can, in principle, implement any desired Pareto optimal allocation as a price-taking equilibrium. We also discuss the practical limitations of this result.

In Section 16.E, we introduce the problem of maximizing a *social welfare function* and relate it to the Pareto optimality concept. We uncover a close formal relationship between these two notions of welfare optimality.

Section 16.F reexamines the Pareto optimality concept and associated results by making differentiability assumptions and analyzing first-order conditions. There we see how equilibrium prices can be interpreted as the Lagrange multipliers, or shadow prices, that arise in the associated Pareto optimality problem.

Section 16.G discusses several applications of the concepts and results previously developed. We first present some examples that rely on particular interpretations of the L abstract commodities; one of them concerns the case of *public goods*. We then consider an application of our results to a world with nonconvex production sets, which leads to a brief exposition of the theory of *marginal cost pricing*.

Appendix A deals with some technical issues concerning the boundedness of the set of feasible allocations and the existence of Pareto optima.

Classical accounts of the material at the heart of this chapter are given by Koopmans (1957), Debreu (1959), and Arrow and Hahn (1971).

16.B The Basic Model and Definitions

In this chapter, we study an economy composed of $I > 0$ consumers and $J > 0$ firms in which there are L commodities. These L commodities can be given many possible interpretations; we discuss some examples in Section 16.G.

Each consumer $i = 1, \dots, I$ is characterized by a consumption set $X_i \subset \mathbb{R}^L$ and a preference relation $\succsim_i$ defined on X_i . We assume that these preferences are rational (i.e., complete and transitive). Chapters 1 to 3 provide an extensive discussion of these concepts.

Each firm $j = 1, \dots, J$ is characterized by a technology, or production set, $Y_j \subset \mathbb{R}^L$. We assume that every Y_j is nonempty and closed. See Chapter 5 for a discussion of production sets and their properties.

The initial resources of commodities in the economy—that is, the economy's *endowments*—are given to us by a vector $\bar{\omega} = (\bar{\omega}_1, \dots, \bar{\omega}_L) \in \mathbb{R}^L$.

Thus, the basic data on preferences, technologies, and resources for this economy are summarized by $(\{(X_i, \succsim_i)\}_{i=1}^I, \{Y_j\}_{j=1}^J, \bar{\omega})$.

The Edgeworth box pure exchange economy discussed in Section 15.B, for example, corresponds to the case in which $L = 2$, $I = 2$, $X_1 = X_2 = \mathbb{R}_+^L$, $J = 1$, and $Y_1 = -\mathbb{R}_+^L$ (the disposal technology). More generally, we say that an economy is a *pure exchange economy* if its only technological possibility is that of free disposal, that is, if $Y_j = -\mathbb{R}_+^L$ for all $j = 1, \dots, J$.

Definition 16.B.1: An *allocation* $(x, y) = (x_1, \dots, x_I, y_1, \dots, y_J)$ is a specification of a consumption vector $x_i \in X_i$ for each consumer $i = 1, \dots, I$ and a production vector $y_j \in Y_j$ for each firm $j = 1, \dots, J$. An allocation (x, y) is *feasible* if

$\sum_i x_{\ell i} = \bar{\omega}_\ell + \sum_j y_{\ell j}$ for every commodity ℓ . That is, if

$$\sum_i x_i = \bar{\omega} + \sum_j y_j. \quad (16.B.1)$$

We denote the set of feasible allocations by

$$A = \{(x, y) \in X_1 \times \cdots \times X_I \times Y_1 \times \cdots \times Y_J : \sum_i x_i = \bar{\omega} + \sum_j y_j\} \subset \mathbb{R}^{L(I+J)}.$$

The notion of a socially desirable outcome that we focus on is that of a *Pareto optimal* allocation.

Definition 16.B.2: A feasible allocation (x, y) is *Pareto optimal* (or *Pareto efficient*) if there is no other allocation $(x', y') \in A$ that *Pareto dominates* it, that is, if there is no feasible allocation (x', y') such that $x'_i \succeq_i x_i$ for all i and $x'_i \succ_i x_i$ for some i .

An allocation is Pareto optimal if there is no waste: It is impossible to make any consumer strictly better off without making some other consumer worse off. Note that the Pareto optimality concept does not concern itself with distributional issues. For example, in a pure exchange economy, an allocation that gives all of society's endowments to one consumer who has strongly monotone preferences is necessarily Pareto optimal.

In Appendix A, we provide conditions on the primitives of the economy implying that the set of feasible allocations A is nonempty, closed, and bounded and that Pareto optimal allocations exist.

Private Ownership Economies

Throughout Part IV, we study the properties of competitive *private ownership economies*. In such economies, every good is traded in a market at publicly known prices that consumers and firms take as unaffected by their own actions. Consumers trade in the marketplace to maximize their well-being, and firms produce and trade to maximize profits. The wealth of consumers is derived from individual endowments of commodities and from ownership claims (*shares*) to the profits of the firms, which are therefore thought of as being owned by consumers.¹

Formally, consumer i has an initial endowment vector of commodities $\omega_i \in \mathbb{R}^L$ and a claim to a share $\theta_{ij} \in [0, 1]$ of the profits of firm j (where $\bar{\omega} = \sum_i \omega_i$ and $\sum_i \theta_{ij} = 1$ for every firm j). Thus, the basic preference, technological, resource, and ownership data of a private ownership economy are summarized by $(\{(X_i, \succeq_i)\}_{i=1}^I, \{Y_j\}_{j=1}^J, \{(\omega_i, \theta_{i1}, \dots, \theta_{iJ})\}_{i=1}^I)$.

The notion of a price-taking equilibrium for a competitive private ownership economy is that of a *Walrasian equilibrium*.

Definition 16.B.3: Given a private ownership economy specified by $(\{(X_i, \succeq_i)\}_{i=1}^I, \{Y_j\}_{j=1}^J, \{(\omega_i, \theta_{i1}, \dots, \theta_{iJ})\}_{i=1}^I)$, an allocation (x^*, y^*) and a price vector $p = (p_1, \dots, p_L)$ constitute a *Walrasian* (or *competitive*) *equilibrium* if:

(i) For every j , y_j^* maximizes profits in Y_j ; that is,

$$p \cdot y_j \leq p \cdot y_j^* \quad \text{for all } y_j \in Y_j.$$

1. Recall from Section 5.G that, under our present assumptions, the consumer-owners of a firm are unanimously in favor of the objective of profit maximization.

(ii) For every i , x_i^* is maximal for $\succeq_i$ in the budget set

$$\{x_i \in X_i: p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*\}^2$$

$$(iii) \sum_i x_i^* = \bar{\omega} + \sum_j y_j^*.$$

Condition (i) of Definition 16.B.3 says that at a Walrasian equilibrium, firms are maximizing their profits given the equilibrium prices p . The logic of profit maximization is examined extensively in Chapter 5. Condition (ii) says that consumers are maximizing their well-being given, first, the equilibrium prices and, second, the wealth derived from their holdings of commodities and from their shares of profits. See Chapter 3 for extensive discussion of preference maximization. Finally, condition (iii) says that markets must clear at an equilibrium; that is, all consumers and firms must be able to achieve their desired trades at the going market prices.

Price Equilibria with Transfers

The aim of this chapter is to relate the idea of Pareto optimality to supportability by means of price-taking behavior. To this end, it is useful to introduce a notion of equilibrium that allows for a more general determination of consumers' wealth levels than that in a private ownership economy. By way of motivation, we can imagine a situation where a social planner is able to carry out (lump-sum) redistributions of wealth, and where society's aggregate wealth can therefore be redistributed among consumers in any desired manner.

Definition 16.B.4: Given an economy specified by $(\{(X_i, \succeq_i)\}_{i=1}^I, \{Y_j\}_{j=1}^J, \bar{\omega})$ an allocation (x^*, y^*) and a price vector $p = (p_1, \dots, p_L)$ constitute a *price equilibrium with transfers* if there is an assignment of wealth levels $(w_1, \dots, w_I)$ with $\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$ such that

(i) For every j , y_j^* maximizes profits in Y_j ; that is,

$$p \cdot y_j \leq p \cdot y_j^* \quad \text{for all } y_j \in Y_j.$$

(ii) For every i , x_i^* is maximal for $\succeq_i$ in the budget set

$$\{x_i \in X_i: p \cdot x_i \leq w_i\}.$$

$$(iii) \sum_i x_i^* = \bar{\omega} + \sum_j y_j^*.$$

The concept of a price equilibrium with transfers requires only that there be *some* wealth distribution such that allocation (x^*, y^*) and price vector $p \in \mathbb{R}^L$ constitute an equilibrium. It captures the idea of price-taking market behavior without any supposition about the determination of consumers' wealth levels. Note that a Walrasian equilibrium is a special case of an equilibrium with transfers. It amounts to the case in which, for every i , consumer i 's wealth level is determined by the initial endowment vector ω_i and by the profit shares $(\theta_{i1}, \dots, \theta_{iJ})$ without any further wealth transfers, that is, where $w_i = p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$ for all $i = 1, \dots, I$.

2. The terminology " x_i is maximal for $\succeq_i$ in set B " means that x_i is a preference-maximizing choice for consumer i in the set B ; that is, $x_i \in B$ and $x_i \succeq_i x'_i$ for all $x'_i \in B$.

16.C The First Fundamental Theorem of Welfare Economics

The first fundamental theorem of welfare economics states conditions under which any price equilibrium with transfers, and in particular any Walrasian equilibrium, is a Pareto optimum. For competitive market economies, it provides a formal and very general confirmation of Adam Smith's asserted "invisible hand" property of the market. A single, very weak assumption, the *local nonsatiation of preferences* (see Section 3.B), is all that is required for the result. Notably, we need not appeal to any convexity assumption whatsoever.

Recall the definition of locally nonsatiated preferences from Section 3.B (Definition 3.B.3).

Definition 16.C.1: The preference relation $\succsim_i$ on the consumption set X_i is *locally nonsatiated* if for every $x_i \in X_i$ and every $\varepsilon > 0$, there is an $x'_i \in X_i$ such that $\|x'_i - x_i\| \leq \varepsilon$ and $x'_i \succ_i x_i$.

Intuitively, the local nonsatiation condition will be satisfied if there are some desirable commodities. Note also a significant implication of the condition: if $\succsim_i$ is continuous and locally nonsatiated, then any closed consumption set X_i must be unbounded. Otherwise, there would by necessity exist a global (hence, local) satiation point (see Exercise 16.C.1).

Proposition 16.C.1: (*First Fundamental Theorem of Welfare Economics*) If preferences are locally nonsatiated, and if (x^*, y^*, p) is a price equilibrium with transfers, then the allocation (x^*, y^*) is Pareto optimal. In particular, any Walrasian equilibrium allocation is Pareto optimal.

Proof: Suppose that (x^*, y^*, p) is a price equilibrium with transfers and that the associated wealth levels are $(w_1, \dots, w_I)$. Recall that $\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$.

The preference maximization part of the definition of a price equilibrium with transfers [i.e., part (ii) of Definition 16.B.4] implies that

$$\text{If } x_i \succ_i x_i^* \quad \text{then } p \cdot x_i > w_i. \quad (16.C.1)$$

That is, anything that is strictly preferred by consumer i to x_i^* must be unaffordable to her. The significance of the local nonsatiation condition for the purpose at hand is that with it (16.C.1) implies an additional property:

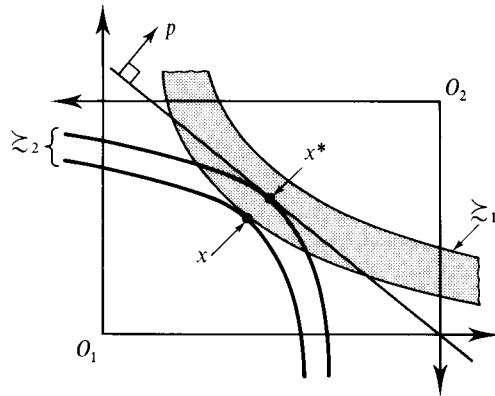
$$\text{If } x_i \succeq_i x_i^* \quad \text{then } p \cdot x_i \geq w_i. \quad (16.C.2)$$

That is, anything that is at least as good as x_i^* is at best just affordable. This property is easily verified (you are asked to do so in Exercise 16.C.2).

Now consider an allocation (x, y) that Pareto dominates (x^*, y^*) . That is, $x_i \succeq_i x_i^*$ for all i and $x_i \succ_i x_i^*$ for some i . By (16.C.2), we must have $p \cdot x_i \geq w_i$ for all i , and by (16.C.1) $p \cdot x_i > w_i$ for some i . Hence,

$$\sum_i p \cdot x_i > \sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*.$$

Moreover, because y_j^* is profit maximizing for firm j at price vector p ,

**Figure 16.C.1**

A price equilibrium with transfers that is not a Pareto optimum

we have $p \cdot \bar{\omega} + \sum_j p \cdot y_j^* \geq p \cdot \bar{\omega} + \sum_j p \cdot y_j$. Thus,

$$\sum_i p \cdot x_i > p \cdot \bar{\omega} + \sum_j p \cdot y_j. \quad (16.C.3)$$

But then (x, y) cannot be feasible. Indeed, $\sum_i x_i = \bar{\omega} + \sum_j y_j$ implies $\sum_i p \cdot x_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j$, which contradicts (16.C.3). We conclude that the equilibrium allocation (x^*, y^*) must be Pareto optimal. ■

The central idea in the proof of Proposition 16.C.1 can be put as follows: At any feasible allocation (x, y) , the total cost of the consumption bundles $(x_1, \dots, x_I)$, evaluated at prices p , must be equal to the social wealth at those prices, $p \cdot \bar{\omega} + \sum_j p \cdot y_j$. Moreover, because preferences are locally nonsatiated, if (x, y) Pareto dominates (x^*, y^*) then the total cost of consumption bundles $(x_1, \dots, x_I)$ at prices p , and therefore the social wealth at those prices, must exceed the total cost of the equilibrium consumption allocation $p \cdot (\sum_i x_i^*) = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$. But by the profit-maximization of Definition 16.B.4, there are no technologically feasible production levels that attain a value of social wealth at prices p in excess of $p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$.

The importance of the nonsatiation assumption for the result can be seen in Figure 16.C.1, which depicts an Edgeworth box where local nonsatiation fails for consumer 1 (note that consumer 1's indifference "curve" is thick) and where the allocation x^* , a price equilibrium for the price vector $p = (p_1, p_2)$ (you should verify this), is not Pareto optimal. Consumer 1 is indifferent about a move to allocation x , and consumer 2, having strongly monotone preferences, is strictly better off. (See Exercise 16.C.3 for a first welfare theorem compatible with satiation.)

Two points about Proposition 16.C.1 should be noted. First, although the result may appear to follow from very weak hypotheses, our theoretical structure already incorporates two strong assumptions: *universal price quoting of commodities* (market completeness) and *price taking* by economic agents. In Part III, we studied a number of circumstances (externalities, market power, and asymmetric information) in which these conditions are not satisfied and market equilibria fail to be Pareto optimal. Second, the first welfare theorem is entirely silent about the desirability of the equilibrium allocation from a distributional standpoint. In Section 16.D, we study the second fundamental theorem of welfare economics. That result, a partial converse to the first welfare theorem, gives us conditions under which any desired distributional aims can be achieved through the use of competitive (price-taking) markets.

16.D The Second Fundamental Theorem of Welfare Economics

The second fundamental welfare theorem gives conditions under which a Pareto optimum allocation can be supported as a price equilibrium with transfers. It is a converse of the first welfare theorem in the sense that it tells us that, under its assumptions, we can achieve any desired Pareto optimal allocation as a market-based equilibrium using an appropriate lump-sum wealth distribution scheme.

The second welfare theorem is more delicate than the first, and its validity requires additional assumptions. To see this, reconsider some of the examples discussed in Chapter 15. In Figure 15.C.3 (a) we saw that in a one-consumer, one-firm economy a Pareto optimal allocation may not be supportable as an equilibrium if the firm's technology is not convex. Figure 15.B.14 depicted a similar failure in a two-consumer Edgeworth box economy where the preferences of one consumer were not convex. Both figures suggest that the assumption of convexity will have to play a central role in establishing the second welfare theorem. Notice that convexity was not appealed to in any way for the first welfare theorem in Section 16.C.

The Edgeworth box of Figure 15.B.10(a) illustrates a different type of failure of supportability by means of prices. In that figure, both consumers have convex preferences, but the Pareto optimal allocation (ω_1, ω_2) cannot be supported as a price equilibrium with transfers; ω_2 is an optimal demand for consumer 2 for any price vector $p = (p_1, p_2) \geq 0$ when her wealth is $w_2 = p \cdot \omega_2$, but ω_1 is an optimal demand for consumer 1 for *no* price vector $p \geq 0$ and wealth level w_1 .

It is convenient to tackle the problems raised by these two types of examples in two steps. The first step consists of establishing a version of the second fundamental theorem in which the sort of failure arising in Figure 15.B.10(a) is allowed. This is accomplished by defining the concept of a *price quasiequilibrium with transfers*, a weakening of the notion of a price equilibrium with transfers. We prove that if all preferences and technologies are convex, any Pareto optimal allocation can be achieved as a price quasiequilibrium with transfers. The second step consists of giving sufficient conditions for a price quasiequilibrium to be a full-fledged equilibrium. This division of labor is convenient because the first step is very general and isolates the central role of convexity, whereas the assumptions for the second step tend to be more special, often being tailored to the particulars of the model under consideration.

The definition of a quasiequilibrium with transfers, Definition 16.D.1, is identical to Definition 16.B.4 except that the preference maximization condition that anything preferred to x_i^* must cost more than w_i (i.e., "if $x_i \succ_i x_i^*$, then $p \cdot x_i > w_i$ ") is replaced by the weaker requirement that anything preferred to x_i^* cannot cost less than w_i (i.e., "if $x_i \succ_i x_i^*$, then $p \cdot x_i \geq w_i$ ").

Definition 16.D.1: Given an economy specified by $(\{(X_i, \succeq_i)\}_{i=1}^I, \{Y_j\}_{j=1}^J, \bar{\omega})$ an allocation (x^*, y^*) and a price vector $p = (p_1, \dots, p_L) \neq 0$ constitute a *price quasiequilibrium with transfers* if there is an assignment of wealth levels $(w_1, \dots, w_I)$ with $\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$ such that

- (i) For every j , y_j^* maximizes profits in Y_j ; that is,
- $$p \cdot y_j \leq p \cdot y_j^* \quad \text{for all } y_j \in Y_j.$$
- (ii) For every i , if $x_i \succ_i x_i^*$ then $p \cdot x_i \geq w_i$.
- (iii) $\sum_i x_i^* = \bar{\omega} + \sum_j y_j^*$.

Part (ii) of Definition 16.D.1 is implied by the preference maximization condition of the definition of a price equilibrium with transfers [part (ii) of Definition 16.B.4]: If x_i^* is preference maximizing in the set $\{x_i \in X_i: p \cdot x_i \leq w_i\}$, then no $x_i \succ_i x_i^*$ with $p \cdot x_i < w_i$ can exist. Hence, any price equilibrium with transfers is a price quasiequilibrium with transfers. However, as we discuss later in this section, the converse is not true.

Note also that when consumers' preferences are locally nonsatiated, part (ii) of Definition 16.D.1 implies $p \cdot x_i^* \geq w_i$ for every i .³ In addition, from part (iii), we get $\sum_i p \cdot x_i^* = p \cdot \bar{\omega} + \sum_j p \cdot y_j^* = \sum_i w_i$. Therefore, *under the assumption of locally nonsatiated preferences*, which we always make, *we must have $p \cdot x_i^* = w_i$ for every i* . This means that we could just as well not mention the w_i 's explicitly and replace part (ii) of Definition 16.D.1 by

- (ii') If $x_i \succ_i x_i^*$ then $p \cdot x_i \geq p \cdot x_i^*$.

That is, allocation (x^*, y^*) and price vector p constitute a price quasiequilibrium with transfers if and only if conditions (i), (ii'), and (iii) hold.⁴ Moreover, with locally nonsatiated preferences, condition (ii') is equivalent to saying that x_i^* is expenditure minimizing on the set $\{x_i \in X_i: x_i \succeq_i x_i^*\}$ (see Exercise 16.D.1). Thus, our discussion later in this section of the conditions under which a price quasiequilibrium with transfers is a price equilibrium with transfers can be interpreted in the locally nonsatiated case as providing conditions under which expenditure minimization on the set $\{x_i \in X_i: x_i \succeq_i x_i^*\}$ implies preference maximization on the set $\{x_i \in X_i: p \cdot x_i \leq p \cdot x_i^*\} = \{x_i \in X_i: p \cdot x_i \leq w_i\}$. ■

Proposition 16.D.1 states a version of the second fundamental welfare theorem.

Proposition 16.D.1: (*Second Fundamental Theorem of Welfare Economics*) Consider an economy specified by $(\{X_i, \succeq_i\}_{i=1}^I, \{Y_j\}_{j=1}^J, \bar{\omega})$, and suppose that every Y_j is convex and every preference relation $\succeq_i$ is convex [i.e., the set $\{x_i' \in X_i: x_i' \succeq_i x_i\}$ is convex for every $x_i \in X_i$] and locally nonsatiated. Then, for every Pareto optimal allocation (x^*, y^*) , there is a price vector $p = (p_1, \dots, p_L) \neq 0$ such that (x^*, y^*, p) is a price quasiequilibrium with transfers.

Proof: In its essence, the proof is just an application of the separating hyperplane theorem for convex sets (see Section M.G. of the Mathematical Appendix). To facilitate comprehension, we organize the proof into a number of small steps.

3. To see this, observe that if preferences are locally nonsatiated and $p \cdot x_i^* < w_i$, then close to x_i^* there is an x_i with $x_i \succ_i x_i^*$ and $p \cdot x_i < w_i$, contradicting condition (ii) of Definition 16.D.1.

4. A similar observation applies, incidentally, to the definition of price equilibrium with transfers (Definition 16.B.4). If preferences are locally nonsatiated, we get an equivalent definition by not referring explicitly to the w_i 's and replacing part (ii) of the definition by (ii''): If $x_i \succ_i x_i^*$ then $p \cdot x_i > p \cdot x_i^*$. Thus, in this locally nonsatiated case, condition (ii'') says that x_i^* is preference maximizing on $\{x_i \in X_i: p \cdot x_i \leq p \cdot x_i^*\}$.

We begin by defining, for every i , the set V_i of consumptions preferred to x_i^* , that is, $V_i = \{x_i \in X_i: x_i \succ_i x_i^*\} \subset \mathbb{R}^L$. Then define

$$V = \sum_i V_i = \left\{ \sum_i x_i \in \mathbb{R}^L: x_i \in V_i, \dots, x_I \in V_I \right\}$$

and

$$Y = \sum_j Y_j = \left\{ \sum_j y_j \in \mathbb{R}^L: y_j \in Y_j, \dots, y_J \in Y_J \right\}.$$

Thus, V is the set of aggregate consumption bundles that could be split into I individual consumptions, each preferred by its corresponding consumer to x_i^* . The set Y is simply the aggregate production set. Note that the set $Y + \{\bar{\omega}\}$, which geometrically is the aggregate production set with its origin shifted to $\bar{\omega}$, is the set of aggregate bundles producible with the given technology and endowments and usable, in principle, for consumption.

Step 1: Every set V_i is convex. Suppose that $x_i \succ_i x_i^*$ and $x'_i \succ_i x_i^*$. Take $0 \leq \alpha \leq 1$. We want to prove that $\alpha x_i + (1 - \alpha)x'_i \succ_i x_i^*$. Because preferences are complete, we can assume without loss of generality that $x_i \succ_i x'_i$. Therefore, by convexity of preferences, we have $\alpha x_i + (1 - \alpha)x'_i \succ_i x'_i$, which by transitivity yields the desired conclusion: $\alpha x_i + (1 - \alpha)x'_i \succ_i x_i^*$ [recall part (iii) of Proposition 1.B.1].

Step 2: The sets V and $Y + \{\bar{\omega}\}$ are convex. This is just a general, and easy-to-prove, mathematical fact: The sum of any two (and therefore any number of) convex sets is convex.

Step 3: $V \cap (Y + \{\bar{\omega}\}) = \emptyset$. This is a consequence of the Pareto optimality of (x^*, y^*) . If there were a vector both in V and in $Y + \{\bar{\omega}\}$, then this would mean that with the given endowments and technologies it would be possible to produce an aggregate vector that could be used to give every consumer i a consumption bundle that is preferred to x_i^* .

Step 4: There is $p = (p_1, \dots, p_L) \neq 0$ and a number r such that $p \cdot z \geq r$ for every $z \in V$ and $p \cdot z \leq r$ for every $z \in Y + \{\bar{\omega}\}$. This follows directly from the separating hyperplane theorem (see Section M. G. the Mathematical Appendix). It is illustrated in Figure 16.D.1.

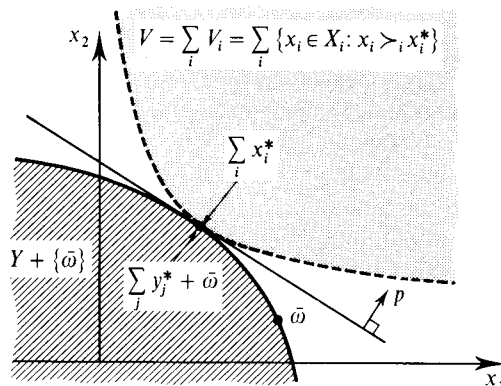


Figure 16.D.1

The separation argument in the proof of the second welfare theorem.

Step 5: If $x_i \succsim_i x_i^*$ for every i then $p \cdot (\sum_i x_i) \geq r$. Suppose that $x_i \succsim_i x_i^*$ for every i . By local nonsatiation, for each consumer i there is a consumption bundle $\hat{x}_i$ arbitrarily close to x_i such that $\hat{x}_i \succ_i x_i$, and therefore $\hat{x}_i \in V_i$. Hence, $\sum_i \hat{x}_i \in V$, and so $p \cdot (\sum_i \hat{x}_i) \geq r$, which, taking the limit as $\hat{x}_i \rightarrow x_i$, gives $p \cdot (\sum_i x_i) \geq r$.⁵

Step 6: $p \cdot (\sum_i x_i^*) = p \cdot (\bar{\omega} + \sum_j y_j^*) = r$. Because of step 5, we have $p \cdot (\sum_i x_i^*) \geq r$. On the other hand, $\sum_i x_i^* = \sum_j y_j^* + \bar{\omega} \in Y + \{\bar{\omega}\}$, and therefore $p \cdot (\sum_i x_i^*) \leq r$. Thus, $p \cdot (\sum_i x_i^*) = r$. Since $\sum_i x_i^* = \bar{\omega} + \sum_j y_j^*$, we also have $p \cdot (\bar{\omega} + \sum_j y_j^*) = r$.

Step 7: For every j , we have $p \cdot y_j \leq p \cdot y_j^*$ for all $y_j \in Y_j$. For any firm j and $y_j \in Y_j$, we have $y_j + \sum_{h \neq j} y_h^* \in Y$. Therefore,

$$p \cdot \left(\bar{\omega} + y_j + \sum_{h \neq j} y_h^* \right) \leq r = p \cdot \left(\bar{\omega} + y_j^* + \sum_{h \neq j} y_h^* \right).$$

Hence, $p \cdot y_j \leq p \cdot y_j^*$.

Step 8: For every i , if $x_i \succ_i x_i^*$, then $p \cdot x_i \geq p \cdot x_i^*$. Consider any $x_i \succ_i x_i^*$. Because of steps 5 and 6, we have

$$p \cdot \left(x_i + \sum_{k \neq i} x_k^* \right) \geq r = p \cdot \left(x_i^* + \sum_{k \neq i} x_k^* \right).$$

Hence, $p \cdot x_i \geq p \cdot x_i^*$.

Step 9: The wealth levels $w_i = p \cdot x_i^*$ for $i = 1, \dots, I$ support (x^*, y^*, p) as a price quasiequilibrium with transfers. Conditions (i) and (ii) of Definition 16.D.1 follow from steps 7 and 8; condition (iii) follows from the feasibility of the Pareto optimal allocation (x^*, y^*) . ■

In Exercise 16.D.2, you are asked to show that the local nonsatiation condition is required in Proposition 16.D.1.

When will a price quasiequilibrium with transfers be a price equilibrium with transfers? The example in Figure 15.B.10(a), reproduced in Figure 16.D.2, indicates that there is indeed a problem. Figure 16.D.2 depicts the quasiequilibrium associated with the Pareto optimal allocation labeled x^* . The unique price vector (normalizing $p_1 = 1$) that supports x^* as a quasiequilibrium allocation is $p = (1, 0)$; the associated wealth levels are $w_1 = p \cdot x_1^* = (1, 0) \cdot (0, x_{21}^*) = 0$ and $w_2 = p \cdot x_2^*$. However, although the consumption bundle x_1^* satisfies part (ii) of Definition 16.D.1 (indeed, $p \cdot x_1 \geq 0 = w_1$ for any $x_1 \geq 0$), it is *not* consumer 1's preference-maximizing bundle in her budget set $\{(x_{11}, x_{21}) \in \mathbb{R}_+^2 : (1, 0) \cdot (x_{11}, x_{21}) \leq 0\} = \{(x_{11}, x_{21}) \in \mathbb{R}_+^2 : x_{11} = 0\}$.

An important feature of the example just discussed, however, is that consumer 1's wealth level at the quasiequilibrium is zero. As we shall see, this is key to the failure of the quasiequilibrium to be an equilibrium. Our next result provides a sufficient condition under which the condition " $x_i \succ_i x_i^*$ implies $p \cdot x_i \geq w_i$ " is equivalent to the preference maximization condition " $x_i \succ_i x_i^*$ implies $p \cdot x_i > w_i$."

5. Geometrically, what we have done here is show that the set $\sum_i \{x_i \in X_i : x_i \succsim_i x_i^*\}$ is contained in the closure of V (see Section M.F of the Mathematical Appendix for this concept), which, in turn, is contained in the half-space $\{v \in \mathbb{R}^L : p \cdot v \geq r\}$.

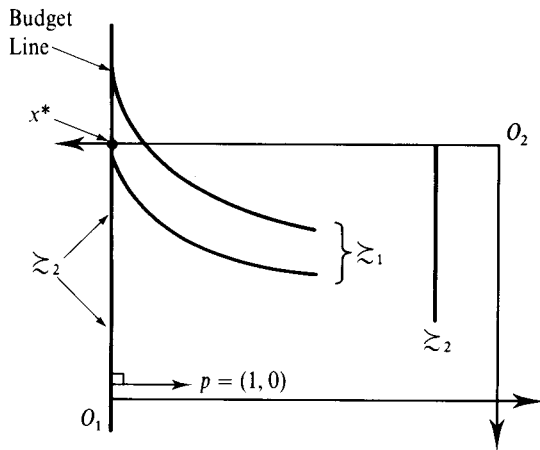


Figure 16.D.2 (left)

A price quasi-equilibrium that is not a price equilibrium.

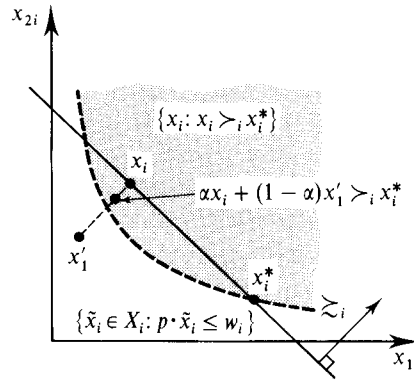


Figure 16.D.3 (right)

Suppose there exists a “cheaper consumption” (an $x'_i \in X_i$ such that $p \cdot x'_i < w_i$). Then if the preferred set does intersect the budget set ($p \cdot x_i \leq w_i$ for some $x_i \succ_i x_i^*$), it follows that the preferred set does intersect the interior of the budget set ($p \cdot x_i < w_i$ for some $x_i \succ_i x_i^*$).

Proposition 16.D.2: Assume that X_i is convex and $\succsim_i$ is continuous. Suppose also that the consumption vector $x_i^* \in X_i$, the price vector p , and the wealth level w_i are such that $x_i \succ_i x_i^*$ implies $p \cdot x_i \geq w_i$. Then, if there is a consumption vector $x'_i \in X_i$ such that $p \cdot x'_i < w_i$ [a *cheaper consumption* for (p, w_i)], it follows that $x_i \succ_i x_i^*$ implies $p \cdot x_i > w_i$.⁶

Proof: The idea of the proof is indicated in Figure 16.D.3 (where we take $p \cdot x_i^* = w_i$ only because this is the leading case; the fact plays no role in the proof). Suppose that, contrary to the assertion of the proposition, there is an $x_i \succ_i x_i^*$ with $p \cdot x_i = w_i$. By the cheaper consumption assumption, there exists an $x'_i \in X_i$ such that $p \cdot x'_i < w_i$. Then for all $\alpha \in [0, 1)$, we have $\alpha x_i + (1 - \alpha)x'_i \in X_i$ and $p \cdot (\alpha x_i + (1 - \alpha)x'_i) < w_i$.⁷ But if α is close enough to 1, the continuity of $\succsim_i$ implies that $\alpha x_i + (1 - \alpha)x'_i \succ_i x_i^*$, which constitutes a contradiction because we have then found a consumption bundle that is preferred to x_i^* and costs less than w_i . ■

Note that in the example of Figure 16.D.2, we have $w_1 = 0$ in the price quasiequilibrium supporting allocation x^* , and so there is no cheaper consumption for (p, w_1) .⁸

As a consequence of Proposition 16.D.2, we have Proposition 16.D.3.

Proposition 16.D.3: Suppose that for every i , X_i is convex, $0 \in X_i$, and $\succsim_i$ is continuous. Then any price quasiequilibrium with transfers that has $(w_1, \dots, w_I) \gg 0$ is a price equilibrium with transfers.

6. If, as in all our applications, $\succsim_i$ is locally nonsatiated and $w_i = p \cdot x_i^*$, then Proposition 16.D.2 offers sufficient conditions for the equivalence of the statements “ x_i^* minimizes expenditure relative to p in the set $\{x_i \in X_i: x_i \succsim_i x_i^*\}$ ” and “ x_i^* is maximal for $\succsim_i$ in the budget set $\{x_i \in X_i: p \cdot x_i \leq p \cdot x_i^*\}$.”

7. A similar argument can be used to show that if X_i is convex and the Walrasian demand function $x_i(p, w_i)$ is well defined, then there is a cheaper consumption for (p, w_i) if and only if there is an x'_i arbitrarily close to $x_i(p, w_i)$ with $p \cdot x'_i < w_i$. In the Appendix A, of Chapter 3 the latter concept was called the *locally cheaper consumption condition*.

8. Note also that Proposition 16.D.2 generalizes the result in Proposition 3.E.1(ii), which assumed local nonsatiation, $w_i = p \cdot x_i^* > 0$, and $X_i = \mathbb{R}_+^L$.

Consider the implications of Proposition 16.D.3 for a pure exchange economy in which $\bar{\omega} \gg 0$ and every consumer has $X_i = \mathbb{R}_+^L$ and continuous, locally nonsatiated preferences. In such an economy, by free disposal and profit maximization, we must have $p \geq 0$ and $p \neq 0$ at any price quasiequilibrium.⁹ Thus, under these assumptions, any price quasiequilibrium with transfers in which $x_i^* \gg 0$ for all i is a price equilibrium with transfers (since then $w_i = p \cdot x_i^* > 0$ for all i). But there is more. Suppose that, in addition, preferences are strongly monotone. Then we must have $p \gg 0$ in any price quasiequilibrium with transfers. To see this, note that $p \geq 0$, $p \neq 0$, and $\bar{\omega} \gg 0$ imply that $\sum_i w_i = p \cdot \bar{\omega} > 0$ and therefore that $w_i > 0$ for some i . But by Proposition 16.D.2, this consumer must then be maximizing her preferences in her budget set $\{x_i \in \mathbb{R}_+^L : p \cdot x_i \leq w_i\}$, which, by strong monotonicity of preferences, cannot occur if prices are not strictly positive. Once we know that we must have $p \gg 0$, we can conclude that any price quasiequilibrium with transfers in this economy is a price equilibrium with transfers: if consumer i 's allocation satisfies $x_i^* \neq 0$, then $p \cdot x_i^* = w_i > 0$ and Proposition 16.D.2 applies. On the other hand, if $x_i^* = 0$, then $w_i = 0$ and the result follows from the fact that $x_i^* = 0$ is the only vector in the set $\{x_i \in \mathbb{R}_+^L : p \cdot x_i \leq 0\}$. (Exercise 16.D.3 asks you to extend the arguments presented in this paragraph to the case of an economy with production.)

The second welfare theorem (combined with Propositions 16.D.2 and 16.D.3) identifies conditions under which any Pareto optimal allocation can be implemented through competitive markets and offers a strong conceptual affirmation of the use of competitive markets, even for dealing with distributional concerns. Yet, it is important to discuss some of the practical limitations on the use of this theoretical result.

The first observation to make is that a planning authority wishing to implement a particular Pareto optimal allocation must be able to insure that the supporting prices $(p_1, \dots, p_L)$ will be taken as given by consumers and firms. If the market structure is such that price-taking behavior would not automatically hold (say, because economic agents are not all of negligible size), then the planning authority must somehow enforce these prices, either by monitoring all transactions or, perhaps, by credibly offering to buy or sell any amount of any good ℓ at price p_ℓ .

A second observation is that the information of a planning authority that wants to use the second welfare theorem must be very good indeed. To begin with, it must have sufficiently good information to identify the Pareto optimal allocation to be implemented and to compute the right supporting price vector. For this purpose, the authority must know, at least, the statistical joint distribution of preferences, endowments, and other relevant characteristics of the agents that actually exist in the economy. However, to implement the correct transfer levels for each consumer, the planning authority must know *more*: it must have the ability to tell who is who by observing each individual's private characteristics (e.g., preferences and endowments) perfectly. Such information is extremely unlikely to be available in practice; as a result, most common transfer schemes fail to be lump-sum schemes. For example, if the planning authority wants to transfer wealth from those who have a great deal of a highly valuable labor skill to those who do not, the only way it may have to tell which consumers are which may be by observing their *actual*

9. Indeed, if we had $p_\ell < 0$, then unboundedly large profits could be generated through disposal of the ℓ th good.

earnings. But if transfers are based on observed earnings, they will cease to be lump-sum in nature. Individuals will recognize that by altering their earnings, they will change their tax burden.

Finally, even if the planning authority observes all the required information, it must actually have the power to enforce the necessary wealth transfers through some tax-and-transfer mechanism that individuals cannot evade.

Because of these informational and enforceability limitations, it is in practice unlikely that extensive lump-sum taxation will be possible.¹⁰ We shall see in Section 18.D that if these types of transfers are not possible, then the second welfare theorem collapses in the sense that, for a typical economy, only a limited range of Pareto optima are supportable by means of prices supplemented by the usual sort of taxation systems. For the typical economy, redistribution schemes are *distortionary*; that is, they trade off distributional aims against Pareto optimality. The analysis of this trade-off is the subject of *second-best* welfare economics, some elements of which are presented in Chapter 22. (Chapter 23 discusses in much more depth what is and what is not implementable by a planning authority who faces informational and enforceability constraints.)

In summary, the second welfare theorem is a very useful *theoretical* reference point. But it is far from a direct prescription for policy practice. On the contrary, by pointing out what is necessary to achieve any desired Pareto optimal allocation, it serves a cautionary purpose.

It is clear from our discussion that convexity plays a central role in the second welfare theorem. But it is a role that deserves a very important qualification. The interpretation of the second welfare theorem is at its strongest when the number of economic agents is large. This is so because the price-taking assumption is then enforced by the market itself (otherwise, it is almost inescapable that there must be some sort of centralized mechanism guaranteeing the fixity of prices). It turns out, however, that if consumers are numerous (in the limit, a continuum), and if the nonconvexities of production sets are bounded in a certain sense, then the assumptions of convex preferences and production sets are *not* required for the second welfare theorem.

To see the idea behind this, it is useful to consider the one-consumer economy depicted in Figure 16.D.4 where, because of nonconvexities, the (trivially Pareto optimal) allocation $x_1 = \bar{\omega}$ cannot be price supported. Suppose, however, that we replicate the economy so that we have *two* consumers and the total endowments are *doubled* to $2\bar{\omega}$. Again, the allocation $x_1 = x_2 = \bar{\omega}$ cannot be price supported, but now *this symmetric allocation is no longer a Pareto optimum*. In Figure 16.D.5, we can see that the asymmetric allocation

$$x'_1 = \bar{\omega} + (1, -1) \quad \text{and} \quad x'_2 = \bar{\omega} + (-1, 1)$$

Pareto dominates $x_1 = x_2 = \bar{\omega}$. It certainly does not follow from this that with just two replicas any allocation that cannot be price supported is not a Pareto optimum. (In Figure 16.D.6, we represent the Edgeworth box associated with Figure 16.D.5. We have drawn it so that the allocation x' is actually Pareto optimal and yet cannot be price supported). However, what

10. Note that the extent of the required lump-sum taxation depends not only on the final wealth levels $(w_1, \dots, w_I)$ but also on the initial situation. For example, if we are in private ownership economy, then the net transfer to consumer i is $w_i - p \cdot \omega_i - \sum_j \theta_{ij} p \cdot y_j^*$ and so depends on the consumer's initial endowments. The Walrasian equilibria correspond to the no-taxation situations; and the farther from the Walrasian wealth levels that we try to go, the larger the required transfer.

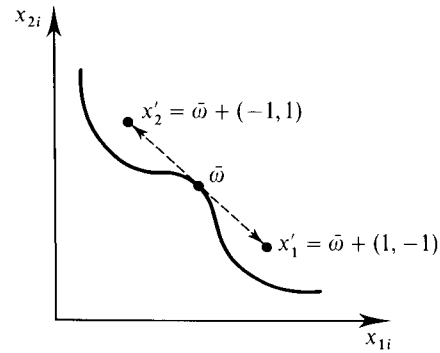
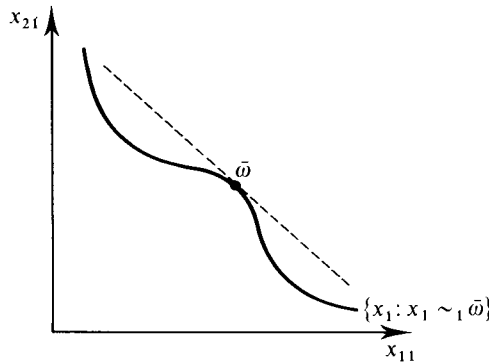


Figure 16.D.4 (top left)
A one-consumer economy (without production) where the initial endowment is not supportable by prices.

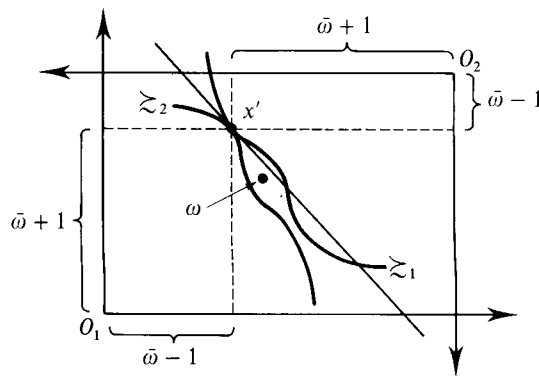


Figure 16.D.5 (top right)
Allocation (x'_1, x'_2) Pareto dominates allocation $(\bar{\omega}, \bar{\omega})$ in a two-consumer replica of the economy in Figure 16.D.4.

Figure 16.D.6 (bottom)
Allocation $x' = (x'_1, x'_2)$ is Pareto optimal but not price supportable.

can be shown is that if the number of replicas is large enough, then any feasible allocation that fails (significantly) to be price supportable can be Pareto dominated, and therefore any Pareto optimum must be (almost) price supportable. (See Exercise 16.D.4 for more on this.)¹¹

16.E Pareto Optimality and Social Welfare Optima

In this section, we discuss the relationship between the Pareto optimality concept and the maximization of a social welfare function (see Section 4.D and Chapter 22 for more on this concept).

Given a family $u_i(\cdot)$ of (continuous) utility functions representing the preferences $\succeq_i$ of the I consumers, we can capture the attainable vectors of utility levels for an economy specified by $((\{X_i, \succeq_i\})_{i=1}^I, \{Y_j\}_{j=1}^J, \bar{\omega})$ by means of the *utility possibility set*:

$$U = \{(u_1, \dots, u_I) \in \mathbb{R}^I : \text{there is a feasible allocation } (x, y) \text{ such that } u_i \leq u_i(x_i) \text{ for } i = 1, \dots, I\}.$$

11. Two facts established in Chapter 17 lend plausibility to this claim. First, in Section 17.1, we show that convexity is not required for the (approximate) existence of a Walrasian equilibrium in a large economy. Second, in Section 17.C, we argue that the second welfare theorem can be rephrased as an assertion of the existence of a Walrasian equilibrium for economies in which endowments are distributed in a particular manner, and it can therefore be seen as implied by the conditions guaranteeing the general existence of Walrasian equilibria.

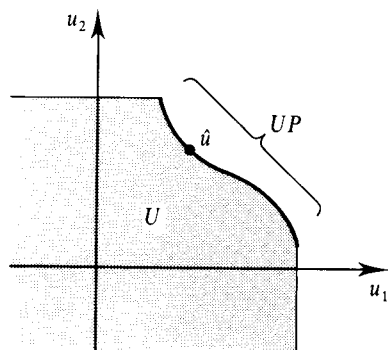


Figure 16.E.1 (left)
The utility possibility set.

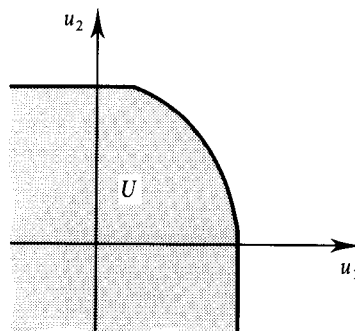


Figure 16.E.2 (right)
A convex utility possibility set.

Figure 16.E.1 depicts this set for a two-consumer economy. (Note that we show $U \subset \mathbb{R}^I$ as a closed set; Appendix A discusses sufficient conditions to guarantee that the set is indeed closed.)

By the definition of Pareto optimality, the utility values of a Pareto optimal allocation must belong to the boundary of the utility possibility set.¹² More precisely, we define the *Pareto frontier* UP , also shown in Figure 16.E.1, by

$$UP = \{(u_1, \dots, u_I) \in U : \text{there is no } (u'_1, \dots, u'_I) \in U \text{ such that } u'_i \geq u_i \text{ for all } i \text{ and } u'_i > u_i \text{ for some } i\}.$$

Proposition 16.E.1 is then intuitive.

Proposition 16.E.1: A feasible allocation $(x, y) = (x_1, \dots, x_I, y_1, \dots, y_J)$ is a Pareto optimum if and only if $(u_1(x_1), \dots, u_I(x_I)) \in UP$.

Proof: If $(u_1(x_1), \dots, u_I(x_I)) \notin UP$, then there is $(u'_1, \dots, u'_I) \in U$ such that $u'_i \geq u_i(x_i)$ for all i and $u'_i > u_i(x_i)$ for some i . But $(u'_1, \dots, u'_I) \in U$ only if there is a feasible allocation (x', y') such that $u_i(x'_i) \geq u'_i$ for all i . It follows then that (x', y') Pareto dominates (x, y) . Conversely, if (x, y) is not a Pareto optimum, then it is Pareto dominated by some feasible (x', y') , which means that $u_i(x'_i) \geq u_i(x_i)$ for all i and $u_i(x'_i) > u_i(x_i)$ for some i . Hence, $(u_1(x_1), \dots, u_I(x_I)) \notin UP$. ■

We also note that if every X_i and every Y_j is convex, and if the utility functions $u_i(\cdot)$ are concave, then the utility possibility set U is convex (see Exercise 16.E.2).¹³ One such utility possibility set is represented in Figure 16.E.2.

Suppose now that society's distributional principles can be summarized in a *social welfare function* $W(u_1, \dots, u_I)$ assigning social utility values to the various possible vectors of utilities for the I consumers. We concentrate here on a particularly simple class of social welfare functions: those that take the *linear* form

$$W(u_1, \dots, u_I) = \sum_i \lambda_i u_i$$

12. However, not every point in the boundary must be Pareto optimal. Go back, for example, to Figure 16.C.1: The utility values associated with x^* belong to the boundary of the utility possibility set because it is impossible to make *both* consumers better off. Yet, x^* is not a Pareto optimum.

13. It can be shown that under a mild technical strengthening of the strict convexity assumption on preferences (essentially the same condition used to guarantee differentiability of the Walrasian demand function in Appendix A of Chapter 3), there are in the family of utility functions $u_i(\cdot)$ that represent $\succeq_i$ some utility functions that are not only quasiconcave but also concave.

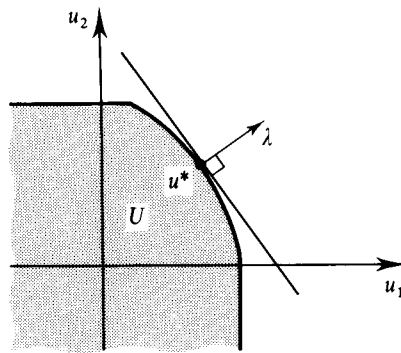


Figure 16.E.3
Maximizing a linear
social welfare function

for some constants $\lambda = (\lambda_1, \dots, \lambda_I)$.¹⁴ Letting $u = (u_1, \dots, u_I)$, we can also write $W(u) = \lambda \cdot u$. Because social welfare should be nondecreasing in the consumer's utility levels, we assume that $\lambda \geq 0$.

Armed with a linear social welfare function, we can select points in the utility possibility set U that maximize our measure of social welfare by solving

$$\text{Max}_{u \in U} \lambda \cdot u. \quad (16.E.1)$$

Figure 16.E.3 depicts the solution to problem (16.E.1). As the figure suggests, we have the result presented in Proposition 16.E.2.

Proposition 16.E.2: If $u^* = (u_1^*, \dots, u_I^*)$ is a solution to the social welfare maximization problem (16.E.1) with $\lambda \gg 0$, then $u^* \in UP$; that is, u^* is the utility vector of a Pareto optimal allocation. Moreover, if the utility possibility set U is convex, then for any $\tilde{u} = (\tilde{u}_1, \dots, \tilde{u}_I) \in UP$, there is a vector of welfare weights $\lambda = (\lambda_1, \dots, \lambda_I) \geq 0$, $\lambda \neq 0$, such that $\lambda \cdot \tilde{u} \geq \lambda \cdot u$ for all $u \in U$, that is, such that $\tilde{u}$ is a solution to the social welfare maximization problem (16.E.1).

Proof: The first part is immediate: if u^* were not Pareto optimal, then there would exist a $u \in U$ with $u \geq u^*$ and $u \neq u^*$; and so because $\lambda \gg 0$, we would have $\lambda \cdot u > \lambda \cdot u^*$.

For the second part, note that if $\tilde{u} \in UP$, then $\tilde{u}$ is in the boundary of U . By the supporting hyperplane theorem (see Section M.G of the Mathematical Appendix), there exists a $\lambda \neq 0$ such that $\lambda \cdot \tilde{u} \geq \lambda \cdot u$ for all $u \in U$. Moreover, since the set U has been constructed so that $U - \mathbb{R}_+^I \subset U$, we must have $\lambda \geq 0$ (indeed, if $\lambda_i < 0$, then by choosing a $u \in U$ with $u_i < 0$ large enough in absolute value, we would have $\lambda \cdot u > \lambda \cdot \tilde{u}$). ■

Proposition 16.E.2 tells us that for economies with convex utility possibility sets, there is a close relation between Pareto optima and linear social welfare optima: Every linear social welfare optimum with weights $\lambda \gg 0$ is Pareto optimal, and every Pareto optimal allocation (and hence, every Walrasian equilibrium) is a social welfare optimum for *some* welfare weights $(\lambda_1, \dots, \lambda_I) \geq 0$.¹⁵

14. See Chapter 22 for a discussion of more general types of social welfare functions.

15. The necessity of allowing for some λ_i to equal zero in the second part of this statement parallels the similar feature encountered in the characterization of efficient production vectors in Proposition 5.F.2.

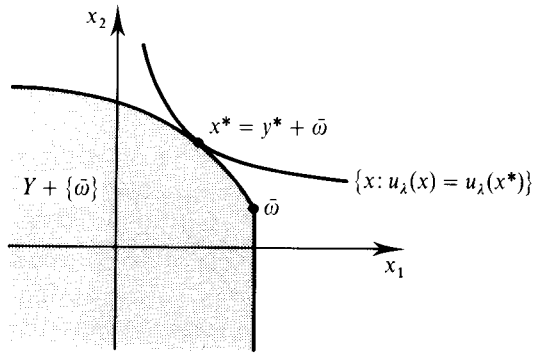


Figure 16.E.4
Maximizing the utility
of a representative
consumer.

As usual, in the absence of convexity of the set U , we cannot be assured that a Pareto optimum can be supported as a maximum of a linear social welfare function. The point $\hat{u}$ in Figure 16.E.1 provides an example where it cannot.

By using the social welfare weights associated with a particular Pareto optimal allocation (perhaps a Walrasian equilibrium), we can view the latter as the welfare optimum in a certain single-consumer, single-firm economy. To see this, let (x^*, y^*) be a Pareto optimal allocation and suppose that $\lambda = (\lambda_1, \dots, \lambda_I) \gg 0$ is a vector of welfare weights supporting U at $(u_1(x_1^*), \dots, u_I(x_I^*))$. Define then a utility function $u_\lambda(\bar{x})$ on aggregate consumption vectors in $X = \sum_i X_i \subset \mathbb{R}^L$ by

$$u_\lambda(\bar{x}) = \text{Max}_{(x_1, \dots, x_I)} \sum_i \lambda_i u_i(x_i) \quad (16.E.2)$$

s.t. $x_i \in X_i$ for all i and $\sum_i x_i = \bar{x}$.

The utility function $u_\lambda(\cdot)$ is the (direct) utility function of a normative representative consumer in the sense discussed in Section 4.D (see, in particular, Exercise 4.D.4). Letting $Y = \sum_j Y_j$ be the aggregate production set, the pair $(\sum_i x_i^*, \sum_j y_j^*)$ is then a solution to the problem

$$\text{Max } u_\lambda(\bar{x})$$

s.t. $\bar{x} = \bar{w} + \bar{y}$, $\bar{x} \in X$, $\bar{y} \in Y$.

This solution is depicted in Figure 16.E.4.

It is important to emphasize, however, that the particular utility function chosen for the representative consumer depends on the weights $(\lambda_1, \dots, \lambda_I)$ and therefore on the Pareto optimal allocation under consideration.

16.F First-Order Conditions for Pareto Optimality

This section intends to be pedagogical. The emphasis is not on minimal assumptions, and its pace is deliberately slow. Making differentiability assumptions, we show how prices and the optimality properties of price-taking behavior emerge naturally from an examination of the first-order conditions associated with Pareto optimality problems. Along the way we redo, and in some aspects also generalize, the analysis in Sections 16.C and 16.D on the two fundamental welfare theorems.¹⁶

16. Early contributions along the line of the discussion in this section are Allais (1953), Lange (1942), and Samuelson (1947).

To begin with, we assume that the consumption set of every consumer is $\mathbb{R}_+^L$ and that preferences are represented by utility functions $u_i(x_i)$ that are twice continuously differentiable and satisfy $\nabla u_i(x_i) \gg 0$ at all x_i (hence, preferences are strongly monotone). We also normalize so that $u_i(0) = 0$.

The production set of firm j takes the form $Y_j = \{y \in \mathbb{R}^L: F_j(y) \leq 0\}$, where $F_j(y) = 0$ defines firm j 's transformation frontier. We assume that $F_j: \mathbb{R}^L \rightarrow \mathbb{R}$ is twice continuously differentiable, that $F_j(0) \leq 0$, and that $\nabla F_j(y_j) = (\partial F_j(y_j)/\partial y_{1j}, \dots, \partial F_j(y_j)/\partial y_{Lj}) \gg 0$ for all $y_j \in \mathbb{R}^L$. The meaning of the last condition is that if $F_j(y_j) = 0$, so that y_j is in the transformation frontier of Y_j , then any attempt to produce more of some output or use less of some input makes the value of $F_j(\cdot)$ positive and pushes us out of Y_j (in other words, y_j is production efficient, in the sense discussed in Section 5.F, in the production set Y_j).¹⁷

Note that, for the moment, no convexity assumptions have been made on preferences or production sets.

The problem of identifying the Pareto optimal allocations for this economy can be reduced to the selection of allocations

$$(x, y) = (x_1, \dots, x_I, y_1, \dots, y_J) \in \mathbb{R}_+^{LI} \times \mathbb{R}^{LJ}$$

that solve the following problem:

$$\begin{aligned} \text{Max} \quad & u_1(x_{11}, \dots, x_{L1}) & (16.F.1) \\ \text{s.t. (1)} \quad & u_i(x_{1i}, \dots, x_{Li}) \geq \bar{u}_i & i = 2, \dots, I \\ & \sum_i x_{\ell i} \leq \bar{\omega}_\ell + \sum_j y_{\ell j} & \ell = 1, \dots, L \\ & F_j(y_{1j}, \dots, y_{Lj}) \leq 0 & j = 1, \dots, J. \end{aligned}$$

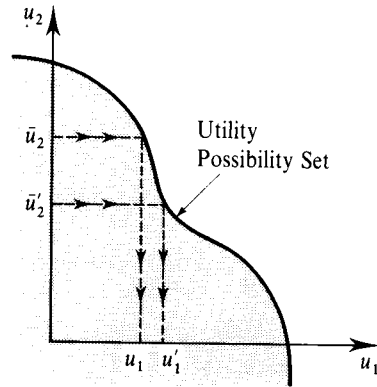
Problem (16.F.1) states the Pareto optimality problem as one of trying to maximize the well-being of consumer 1 subject to meeting certain required utility levels for the other consumers in the economy [constraints (1)] and the resource and technological limitations on what is feasible [constraints (2) and (3), respectively]. By solving problem (16.F.1) for varying required levels of utility for these other consumers $(\bar{u}_2, \dots, \bar{u}_I)$, we can identify all the Pareto optimal allocations for this economy. Indeed, you should pause to convince yourself of this by solving Exercise 16.F.1.

Exercise 16.F.1: Show that any allocation that is a solution to problem (16.F.1) is Pareto optimal and that any Pareto optimal allocation for this economy must be a solution to problem (16.F.1) for *some* choice of utility levels $(\bar{u}_2, \dots, \bar{u}_I)$. [Hint: Use the fact that preferences are strongly monotone.]

Because utility functions are normalized to take nonnegative values, from now on we consider only required utility levels that satisfy $\bar{u}_i \geq 0$ for all i .

The point of Exercise 16.F.1 can be seen by examining the utility possibility set U in Figure 16.F.1. If we fix a required nonnegative utility level for consumer 2, we can locate a point on the frontier of the utility possibility set U by maximizing

17. For expositional convenience, we have taken every $F_j(\cdot)$ to be defined on the entire $\mathbb{R}^L$. A consequence of this (and the assumption that $\nabla F_j(y_j) \gg 0$ for all y_j) is that every commodity is both an input and an output of the production process. Because this is unrealistic, we emphasize that no more than expositional ease is involved here.

**Figure 16.F.1**

Parameterizing the frontier of the utility possibility set when $I = 2$ by the required utility level of consumer 2.

consumer 1's utility subject to the required utility constraint for consumer 2. By varying 2's required utility level, we trace out the set of Pareto optimal points.

Under our assumptions (recall that $\bar{u}_i \geq 0$ for $i \geq 2$), all the constraints of problem (16.F.1) will be binding at a solution. Denote by $(\delta_2, \dots, \delta_I) \geq 0$, $(\mu_1, \dots, \mu_L) \geq 0$, and $(\gamma_1, \dots, \gamma_J) \geq 0$ the multipliers associated with the constraints (1), (2), and (3) of problem (16.F.1), respectively, and define $\delta_1 = 1$. In Exercise 16.F.2, you are asked to verify that the first-order (Kuhn–Tucker) necessary conditions for problem (16.F.1) can be written as follows (all the derivatives, here and elsewhere in this section, are evaluated at the solution):¹⁸

$$x_{\ell i}: \quad \delta_i \frac{\partial u_i}{\partial x_{\ell i}} - \mu_{\ell} \begin{cases} \leq 0 \\ = 0 \text{ if } x_{\ell i} > 0 \end{cases} \quad \text{for all } i, \ell, \quad (16.F.2)$$

$$y_{\ell j}: \quad \mu_{\ell} - \gamma_j \frac{\partial F_j}{\partial y_{\ell j}} = 0 \quad \text{for all } j, \ell. \quad (16.F.3)$$

As is well known from Kuhn–Tucker theory (see Section M.K of the Mathematical Appendix), the value of the multiplier μ_{ℓ} at an optimal solution is exactly equal to the increase in consumer 1's utility derived from a relaxation of the corresponding constraint, that is, from a marginal increase in the available social endowment $\bar{\omega}_{\ell}$ of good ℓ . Thus, the multiplier μ_{ℓ} can be interpreted as the marginal value or “shadow price” (in terms of consumer 1's utility) of good ℓ . The multiplier δ_i , on the other hand, equals the marginal change in consumer 1's utility if we decrease the utility requirement $\bar{u}_i$ that must be met for consumer $i \neq 1$. Condition (16.F.2) therefore says that, at an optimal interior allocation, the increase in the utility of any consumer i from receiving an additional unit of good ℓ , weighted (if $i \neq 1$) by the amount that relaxing consumer i 's utility constraint is worth in terms of raising consumer 1's utility, should be equal to the marginal value μ_{ℓ} of good ℓ .

Similarly, the multiplier γ_j can be interpreted as the marginal benefit from relaxing the j th production constraint or, equivalently, the marginal cost from tightening it.

18. Recall that for expositional ease we are not imposing any boundary constraints on the vectors y_j . We note also that the assumption of strictly positive gradients of the functions $u_i(\cdot)$ and $F_j(\cdot)$ implies that the constraint qualification for the necessity of the Kuhn–Tucker conditions is satisfied. (See Section M.K of the Mathematical Appendix for the specifics of first-order conditions for optimization problems under constraints.)

Hence, $\gamma_j(\partial F_j/\partial y_{\ell j})$ is the marginal cost of increasing $y_{\ell j}$ and thereby effectively tightening the constraint on the net outputs of the other goods. Condition (16.F.3) says, then, that at an optimum this marginal cost is equated, for every j , to the marginal benefit μ_{ℓ} of good ℓ .

If we suppose that we have an interior solution (i.e., $x_i \gg 0$ for all i), then conditions (16.F.2) and (16.F.3) imply that three types of ratio conditions must hold (see Exercise 16.F.3):

$$\frac{\partial u_i/\partial x_{\ell i}}{\partial u_i/\partial x_{\ell' i}} = \frac{\partial u_{i'}/\partial x_{\ell i'}}{\partial u_{i'}/\partial x_{\ell' i'}} \quad \text{for all } i, i', \ell, \ell'. \quad (16.F.4)$$

$$\frac{\partial F_j/\partial y_{\ell j}}{\partial F_j/\partial y_{\ell' j}} = \frac{\partial F_{j'}/\partial y_{\ell j'}}{\partial F_{j'}/\partial y_{\ell' j'}} \quad \text{for all } j, j', \ell, \ell'. \quad (16.F.5)$$

$$\frac{\partial u_i/\partial x_{\ell i}}{\partial u_i/\partial x_{\ell' i}} = \frac{\partial F_j/\partial y_{\ell j}}{\partial F_j/\partial y_{\ell' j}} \quad \text{for all } i, j, \ell, \ell'. \quad (16.F.6)$$

Condition (16.F.4) says that in any Pareto optimal allocation, all consumers' marginal rates of substitution between every pair of goods must be equalized [see Figures 15.B.11(b) and 15.B.12 for an illustration in the two-good, two-consumer case]; condition (16.F.5) says that all firms' marginal rates of transformation between every pair of goods must be equalized [see Figure 15.D.2(b) for an illustration in the two-good, two-firm case]; and condition (16.D.6) says that every consumer's marginal rate of substitution must equal every firm's marginal rate of transformation for all pairs of goods [see Figure 15.C.2 for an illustration in the case of the one-consumer, one-firm model with two goods].

Conditions (16.F.4) to (16.F.6) correspond to three types of efficiency embodied in a Pareto optimal allocation (see Exercise 16.F.4).

(i) *Optimal allocation of available goods across consumers.* Given some aggregate amounts $(\bar{x}_1, \dots, \bar{x}_L)$ of goods available for consumption purposes, we want to distribute them to maximize consumer 1's well-being while meeting the utility requirements $(\bar{u}_2, \dots, \bar{u}_I)$ for consumers 2, $\dots$, I . That is, we want to solve

$$\begin{aligned} \text{Max}_{(x_1, \dots, x_I)} \quad & u_1(x_{11}, \dots, x_{L1}) \\ \text{s.t. (1)} \quad & u_i(x_{1i}, \dots, x_{Li}) \geq \bar{u}_i \quad i = 2, \dots, I \\ & (2) \sum_i x_{\ell i} \leq \bar{x}_{\ell} \quad \ell = 1, \dots, L. \end{aligned} \quad (16.F.7)$$

The first-order conditions for this problem lead to condition (16.F.4).

(ii) *Efficient production across technologies.* The aggregate production vector should be efficient in the sense discussed in Section 5.F. That is, it should be impossible to reassign production plans across individual production sets so as to produce, in the aggregate, more of a particular output (or use less of it as an input) without producing less of another. Focusing, in particular, on the first good, this means that given required total productions $(\bar{y}_2, \dots, \bar{y}_L)$ of the other goods, we want to solve

$$\begin{aligned} \text{Max}_{(y_1, \dots, y_J)} \quad & \sum_j y_{1j} \\ \text{s.t. (1)} \quad & \sum_j y_{\ell j} \geq \bar{y}_{\ell} \quad \ell = 2, \dots, L \\ & (2) F_j(y_j) \leq 0 \quad j = 1, \dots, J. \end{aligned} \quad (16.F.8)$$

The first-order conditions for this problem lead to condition (16.F.5).

(iii) *Optimal aggregate production levels.* We also must have picked aggregate production levels that generate a desirable assortment of commodities available for consumption. Keeping the utility requirements $(\bar{u}_2, \dots, \bar{u}_I)$ fixed, let $u(\bar{x}_1, \dots, \bar{x}_L)$ and $f(\bar{y}_2, \dots, \bar{y}_L)$ denote, respectively, the value functions for problems (16.F.7) and (16.F.8). Then we want to solve

$$\begin{aligned} \text{Max}_{(\bar{y}_1, \dots, \bar{y}_L)} \quad & u(\bar{\omega}_1 + \bar{y}_1, \dots, \bar{\omega}_L + \bar{y}_L) \\ \text{s.t.} \quad & \bar{y}_1 \leq f(\bar{y}_2, \dots, \bar{y}_L). \end{aligned} \quad (16.F.9)$$

The first-order conditions of this problem lead to condition (16.F.6).

To explore the relationship of the first-order conditions (16.F.2) and (16.F.3) to the first and second welfare theorems, we make the further, and substantive, assumption that every $u_i(\cdot)$ is a quasiconcave function (hence, preferences are convex) and that every $F_j(\cdot)$ is a convex function (hence, production sets are convex). The virtue of this assumption is that with it we do not have to worry about second-order conditions; in all the maximization problems to be considered, the first-order necessary conditions are automatically sufficient.

In this differentiable, convex framework, conditions (16.F.2) and (16.F.3) can be used to establish a version of the two welfare theorems. To see this, note first that (x^*, y^*, p) is a price equilibrium with transfers (with associated wealth levels $w_i = p \cdot x_i^*$ for $i = 1, \dots, I$) if and only if the first-order conditions for the budget-constrained utility maximization problems

$$\begin{aligned} \text{Max}_{x_i \geq 0} \quad & u_i(x_i) \\ \text{s.t.} \quad & p \cdot x_i \leq w_i \end{aligned}$$

and the profit maximization problems

$$\begin{aligned} \text{Max}_{y_j} \quad & p \cdot y_j \\ \text{s.t.} \quad & F_j(y_j) \leq 0 \end{aligned}$$

are satisfied. Denoting by α_i and β_j the respective multipliers for the constraints of these problems, the first-order conditions [evaluated at (x^*, y^*)] can be written as follows:

$$x_{\ell i}: \quad \frac{\partial u_i}{\partial x_{\ell i}} - \alpha_i p_{\ell} \begin{cases} \leq 0 \\ = 0 \text{ if } x_{\ell i} > 0 \end{cases} \quad \text{for all } i, \ell, \quad (16.F.10)$$

$$y_{\ell j}: \quad p_{\ell} - \beta_j \frac{\partial F_j}{\partial y_{\ell j}} = 0 \quad \text{for all } i, \ell. \quad (16.F.11)$$

Letting $\mu_{\ell} = p_{\ell}$, $\delta_i = 1/\alpha_i$, and $\gamma_j = \beta_j$, we see that there is an exact correspondence between conditions (16.F.2)–(16.F.3) and (16.F.10)–(16.F.11). Since both sets of conditions are necessary and sufficient for their respective problems, this implies that the allocation (x^*, y^*) is Pareto optimal if and only if it is a price equilibrium with transfers with respect to some price vector $p = (p_1, \dots, p_L)$. Note, moreover, that the equilibrium price p_{ℓ} exactly equals μ_{ℓ} , the marginal value of good ℓ in the Pareto optimality problem.

Suppose that, in addition, every $u_i(\cdot)$ is concave. Then it is also instructive to examine the marginal conditions for the maximization of a linear social welfare

function (see Section 16.E). Consider the problem

$$\begin{aligned} \text{Max}_{x,y} \quad & \sum_i \lambda_i u_i(x_{1i}, \dots, x_{Li}) \\ \text{s.t.} \quad & (1) \quad \sum_i x_{\ell i} \leq \bar{\omega}_\ell + \sum_j y_{\ell j} \quad \ell = 1, \dots, L \\ & (2) \quad F_j(y_{1j}, \dots, y_{Lj}) \leq 0 \quad j = 1, \dots, J. \end{aligned} \quad (16.F.12)$$

where $\lambda_i > 0$ for all i . Letting $(\psi_1, \dots, \psi_L)$ and $(\eta_1, \dots, \eta_J)$ denote the multipliers on constraints (1) and (2), respectively, the necessary and sufficient first-order conditions for this problem can be written as follows:

$$x_{\ell i}: \quad \lambda_i \frac{\partial u_i}{\partial x_{\ell i}} - \psi_\ell \begin{cases} \leq 0 \\ = 0 \text{ if } x_{\ell i} > 0 \end{cases} \quad \text{for all } i, \ell, \quad (16.F.13)$$

$$y_{\ell j}: \quad \psi_\ell - \eta_j \frac{\partial F_j}{\partial y_{\ell j}} = 0 \quad \text{for all } j, \ell. \quad (16.F.14)$$

Note that by letting $\delta_i = \lambda_i/\lambda_1$, $\mu_\ell = \psi_\ell/\lambda_1$, and $\gamma_j = \eta_j/\lambda_1$, we have an exact correspondence between (16.F.2)–(16.F.3) and (16.F.13)–(16.F.14). Therefore, any solution to (16.F.13) and (16.F.14) is a solution to (16.F.2) and (16.F.3) and, hence, a Pareto optimum.¹⁹ Conversely, any Pareto optimum that for some multipliers satisfies (16.F.2) and (16.F.3) is also a solution of (16.F.13) and (16.F.14), and consequently of problem (16.F.12), for an appropriate choice of $\lambda = (\lambda_1, \dots, \lambda_I)$.

It is also enlightening to compare (16.F.13) and (16.F.14) with the first-order conditions (16.F.10) and (16.F.11) for the optimization problems associated with a price equilibrium with transfers. We get an exact correspondence between them by letting $p_\ell = \psi_\ell$, $\alpha_i = 1/\lambda_i$, and $\beta_j = \eta_j$. Once again, the price p_ℓ represents the marginal social value of good ℓ . In addition, note that α_i , which is the marginal utility of wealth for consumer i at prices p and wealth level $w_i = p \cdot x_i^*$, equals the reciprocal of λ_i . Hence, we can draw the conclusion presented in Proposition 16.F.1.

Proposition 16.F.1: Under the assumptions made about the economy [in particular, the concavity of every $u_i(\cdot)$ and the convexity of every $F_j(\cdot)$], every Pareto optimal allocation (and, hence, every price equilibrium with transfers) maximizes a weighted sum of utilities subject to the resource and technological constraints. Moreover, the weight λ_i of the utility of the i th consumer equals the reciprocal of consumer i 's marginal utility of wealth evaluated at the supporting prices and imputed wealth.

16.G Some Applications

In this section, we present some applications of the ideas covered in the previous sections of the chapter. We first discuss three examples that introduce particular interpretations of the commodity space. We then present an extension of the second welfare theorem that relies on a concept of *marginal cost pricing*.

19. Recall that by the concavity–convexity assumptions, (16.F.2)–(16.F.3) and (16.F.13)–(16.F.14) are necessary and sufficient conditions for their respective problems.

Interpretations of the Commodity Space

Up to now we have treated our commodities as abstractly defined objects. This has not been for formalism's sake, but to facilitate a wide applicability of the theory.

It is easy to think of the case in which commodities are distinct, physically tradeable real objects. But there are many other interesting possibilities. The theory presented in the previous sections has proven to be remarkably flexible and subtle in the interpretations that can be given to the commodities, consumption sets, preferences, and production sets.

Two important examples, *commodities contingent on the state of the world* and *dated commodities*, are discussed extensively in Chapters 19 and 20, respectively. For the sake of completeness, we devote a few words to contingent commodities in this section. We then briefly discuss two other examples: *occupational choice* and *public goods*.

Example 16.G.1: Contingent Commodities. An interesting use of artificial commodities appears in the area of general equilibrium under uncertainty. A full formal description is presented in Chapter 19, but the basic idea can be conveyed here. The usefulness of a commodity may depend on uncertain, external circumstances. For example, medical care is much more important if one is ill than if one is healthy. To ensure an efficient allocation of resources, we have to make sure not only that the right commodities are delivered to the right people but also that they are allocated under the right circumstances, that is, according to the realization of the uncertain external states. To model this type of resource allocation problem, we can use the concept of a *contingent commodity*. A commodity such as medical care can be subdivided into many different "artificial commodities," each of which has the interpretation "medical care is provided under circumstance s ." For example, suppose that there are I consumers in the economy, each of whom may turn out ex post to be either "sick" or "healthy." A consumer's need for medical care depends, of course, on her state of health. From an economy-wide perspective, there are then 2^I different states of nature, each corresponding to a different configuration of ill health across the population. We can therefore imagine 2^I different commodities called "medical care," one for each of these configurations. A consumer buying "medical care in state s " receives care when state s occurs.²⁰

One of the strengths of general equilibrium theory is its ability to deal easily with an arbitrary number of commodities. There are very few results that depend on the number of commodities, and none of them is of general interest. Therefore, even though it seems difficult to conceive of a very large number of markets for a very large number of contingent commodities, all the welfare propositions that we have developed turn out to be easily applicable to this uncertainty setting (to be sure, we are taking a theoretical, rather than a practical perspective here). In Chapter 19, we discuss these points in more detail. ■

20. Thus, to purchase medical care when she is sick, the consumer actually buys a large number of different "contingent medical care commodities" (in fact, 2^{I-1} of them).

Example 16.G.2: Occupational Choice Suppose that every individual could, in principle, work either as a classics scholar or as an economics professor. But not all individuals are equally good at both things. A way to capture the different comparative advantage is to assume that for every individual i , there is an $\alpha_i \geq 0$ measuring how many “effective hours of economics professorial services” it takes to produce “an effective hour of classical scholarship.” A relatively low α_i indicates comparative advantage in classical scholarship. Suppose also that every individual i has an amount $\bar{T}_i$ of professorial hours that she can supply; we assume that 1 professorial hour can produce 1 effective hour of economics professorial services or $1/\alpha_i$ effective hours of classical scholarship by individual i . There is a single consumption good on which the individual i can spend her earnings.

It is important to be able to imbed this problem in our formal structure because we certainly want to be able to analyze how, for example, competitive labor markets will perform when individuals have *occupational choices* as well as choices about *how much* labor to supply.

This is how it can be done (it is not the only possible way): suppose we list consumption and effective hours supplied as a three-dimensional vector (c_i, t_{ci}, t_{ei}) , where c_i is individual i 's consumption and $t_{ci} \leq 0$ and $t_{ei} \leq 0$ are the effective hours spent working as a classics scholar and as an economics professor, respectively. Because the latter two quantities are supplies—that is, services offered by the individual to the market—we follow the convention of measuring them as negative numbers. We can then define the consumption set of individual i as

$$X_i = \{(c_i, t_{ci}, t_{ei}) : c_i \geq 0, t_{ci} \leq 0, t_{ei} \leq 0, \bar{T}_i + t_{ei} + \alpha_i t_{ci} \geq 0\}.$$

One should interpret the nonpositivity constraints as the inability to consume labor services. The amount $\bar{T}_i + t_{ei} + \alpha_i t_{ci}$ is the time available for leisure activities. Preferences are defined on X_i . Because the consumption good is desirable, the local nonsatiation condition is satisfied. The assumption that preferences are continuous and convex is also natural. We can complete the model by having a concave production function $f(z_c, z_e)$ that transforms input combinations (z_c, z_e) of effective hours of classics and economics scholarship, respectively, into the consumption good.

We now have a complete general equilibrium system to which we can pose a number of interesting questions: If the occupational choice is directed by a competitive (i.e., price-taking) market system, will the outcome result in an efficient exploitation of comparative advantage? Conversely, can every efficient arrangement of occupations be sustained by a market system (supplemented perhaps by lump-sum transfers)? The results of Sections 16.C and 16.D tell us that the answer to both questions is in the affirmative. ■

Example 16.G.3: Public Goods. The notion of a “public good” and the more general concept of an “externality” were discussed in Chapter 11, where we also introduced the key idea of “personalized” prices.²¹ Consequently, we can be rather brief here. (The basic references on public goods were also given in Chapter 11.)

21. In fact, the current discussion is more general than the one in Chapter 11 because there we restricted ourselves to a quasilinear setting.

Suppose there are I consumers and two commodities; a “private” good, say labor, and a public good (the theory presented extends without essential modification to any number of private or public goods). A *private good* is like the goods discussed up to now: A unit of the good can be consumed only once, so if consumer i consumes it then it is unavailable for consumption by others. We let x_{1i} denote consumer i ’s consumption of the private good. But in the case of a (pure) *public good*, its consumption by consumer i does not prevent its availability to other consumers. Thus, if x_2 is the total amount of public good produced, then x_2 can be made available at no extra cost to *every* consumer.

We assume that every consumer i has the consumption set $\mathbb{R}_+^2$ and continuous, convex, locally nonsatiated preferences $\succeq_i$ defined on pairs (x_{1i}, x_2) . The model is completed by having some amount $\bar{\omega}_1$ of the private good as the initial total endowment (there is no endowment of the public good) and a firm that transforms amounts $z \in \mathbb{R}_+$ of the private good into the public good by means of an increasing, concave production function $f(z)$.

An allocation $((x_{11}, \dots, x_{1I}, x_2), (q, z)) \geq 0$ is feasible if

$$q \leq f(z), \quad \sum_i x_{1i} + z = \bar{\omega}_1, \quad \text{and} \quad q = x_2.$$

It is Pareto optimal if there is no other feasible allocation $((x'_{11}, \dots, x'_{1I}, x'_2), (q', z'))$ such that $(x'_{1i}, x'_2) \succeq_i (x_{1i}, x_2)$ for all i and $(x'_{1i}, x'_2) \succ_i (x_{1i}, x_2)$ for some i .

We now describe this model in an artificial but equivalent way, with the advantage, that, formally speaking, it reduces the public commodities to private ones and therefore makes the results of Sections 16.C and 16.D applicable. The “trick” is to define a personalized commodity x_{2i} for every consumer i , to be interpreted as “commodity 2 as received by the i th consumer.” Formally, consumer i cares only about good 1 and the i th personalized commodity. We therefore denote her consumption bundle by $x_i = (x_{1i}, x_{2i})$. The single firm is now viewed as producing a *joint* bundle of personalized commodities with a technology that produces these commodities in fixed proportions. Formally, its (convex) production set is

$$Y = \{(-z, q_1, \dots, q_I) \in \mathbb{R}_+^{I+1} : z \geq 0 \text{ and } q_1 = \dots = q_I = q \leq f(z)\}.$$

With this reinterpretation, the model fits into the structure analyzed throughout the chapter. A price equilibrium with transfers for this artificial economy is known as a *Lindahl equilibrium*.²²

Definition 16.G.1: A *Lindahl equilibrium* for the public goods economy is a price equilibrium with transfers for the artificial economy with personalized commodities. That is, an allocation $(x_1^*, \dots, x_I^*, q^*, z^*) \in \mathbb{R}^{2I} \times \mathbb{R} \times \mathbb{R}$ and a price system $(p_1, p_{21}, \dots, p_{2I}) \in \mathbb{R}^{I+1}$ constitute a Lindahl equilibrium if there is a set of wealth levels $(w_1, \dots, w_I)$ satisfying $\sum_i w_i = \sum_i p_1 x_{1i}^* + (\sum_i p_{2i}) q^* - p_1 z^*$ and such that:

- (i) $q^* \leq f(z^*)$ and $(\sum_i p_{2i}) q^* - p_1 z^* \geq (\sum_i p_{2i}) q - p_1 z$ for all (q, z) with $z \geq 0$ and $q \leq f(z)$.

22. More properly, we should say a *Lindahl equilibrium with transfers*.

- (ii) For every i , $x_i^* = (x_{1i}^*, x_{2i}^*)$ is maximal for $\succsim_i$ in the set $\{(x_{1i}, x_{2i}) \in X_i: p_1 x_{1i} + p_{2i} x_{2i} \leq w_i\}$.
- (iii) $\sum_i x_{1i}^* + z^* = \bar{\omega}_1$ and $x_{2i}^* = q^*$ for every i .

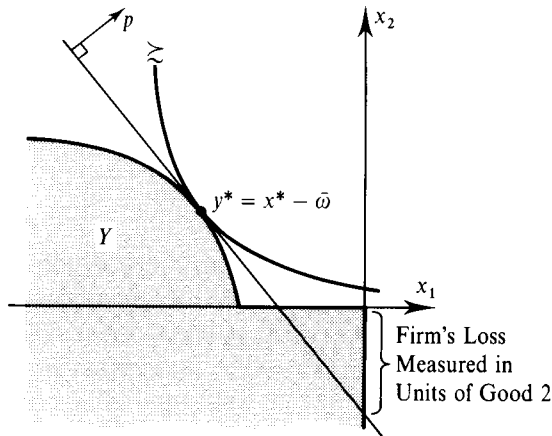
The first and second fundamental welfare theorems tell us that every Lindahl equilibrium is Pareto optimal and every Pareto optimal allocation can be implemented using a Lindahl equilibrium (with appropriate wealth transfers and, perhaps, with the usual quasiequilibrium qualification).²³ There is an important caveat, however: unlike economies where with large numbers of agents each agent becomes small relative to the size of the market, in markets for personalized goods, each consumer is necessarily large with respect to the market in her personalized good. As we multiply the number of consumers, we also multiply, as a matter of definition, the number of commodities. As a result, it is very unlikely that the critical assumption of price taking will be satisfied. Thus, the descriptive content of this equilibrium concept is low.

Nevertheless, the second welfare theorem may still be of some interest. In particular, it tells us that if the planning authority has a means to enforce the prices, then we have a mechanism involving voluntary purchases of the public good that achieves the desired Pareto optimal allocation. Even for this purpose, however, further difficulties arise that are inherent to the public goods setting: First, to calculate the personalized supporting prices, statistical information (e.g. information on the distribution of preferences across the economy) will not do; the fact that prices are personalized means that personal, private information is required. This information may be difficult to get because individuals will often not have incentives to reveal this information truthfully (see Chapters 11 and 23 for more on this issue). Second, for a personalized market voluntary mechanism to work, individuals must expect to receive precisely the amount of public good they purchase. This requires that the public good be *excludable*; that is, there must be some procedure to deny total or partial use of the public good to anyone who does not pay for it. In many cases, such exclusion is difficult, if not impossible (consider, for example, national defense). ■

Nonconvex Production Technologies and Marginal Cost Pricing

The second welfare theorem runs into difficulties in the presence of nonconvex production sets (in this section, we do not question the assumption of convexity on the consumption side). In the first place, large nonconvexities caused by the presence of fixed costs or extensive increasing returns lead to a world of a small number of large firms (in the limit, production efficiency may require a single firm, a so-called “natural monopoly”), making the assumption of price taking less plausible. Yet, even

23. Suppose that the production function $f(\cdot)$ is differentiable and the indifference curves are smooth, and consider a Lindahl equilibrium that is interior. Then preference maximization implies that $p_{2i}/p_1 = -MRS_{21}^i$, where MRS_{21}^i is consumer i 's marginal rate of substitution of good 2 for good 1. On the other hand, profit maximization entails $\sum_i p_{2i}/p_1 = -MRT_{21}$, where MRT_{21} is the firm's marginal rate of transformation of good 2 for good 1 (the marginal cost of output in terms of input). Hence, in any Lindahl equilibrium, we must have $\sum_i MRS_{21}^i = MRT_{21}$, which is exactly the Samuelson optimality condition for a public good (see Section 11.G for its derivation in the case of quasilinear preferences).

**Figure 16.G.1**

The firm incurs a loss at the prices that locally support the Pareto optimal allocation.

if price taking can somehow be relied on (perhaps because a planning authority can enforce prices), it may still be impossible to support a given Pareto optimal allocation. Examples are provided by Figures 15.C.3(a) and 16.G.1. In Figure 16.G.1, at the only relative prices that could support the production y^* locally, the firm sustains a loss and would rather avoid it by shutting down. In Figure 15.C.3(a), on the other hand, not even local profit maximization can be guaranteed (see the discussion in Section 15.C).

Although nonconvexities may prevent us from supporting the Pareto optimal production allocation as a profit-maximizing choice, under the differentiability assumptions of Section 16.F we can use the first-order necessary conditions derived there to formulate a weaker result that parallels the second welfare theorem (see Exercise 16.G.1).

Proposition 16.G.1: Suppose that the basic assumptions of Section 16.F hold²⁴ and that, in addition, all consumers have convex preferences (so utility functions are quasiconcave). If (x^*, y^*) is Pareto optimal, then there exists a price vector $p = (p_1, \dots, p_L)$ and wealth levels $(w_1, \dots, w_I)$ with $\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$ such that:

(i) For any firm j , we have

$$p = \gamma_j \nabla F_j(y_j^*) \quad \text{for some } \gamma_j > 0. \quad (16.G.1)$$

(ii) For any i , x_i^* is maximal for $\succsim_i$ in the budget set

$$\{x_i \in X_i : p \cdot x_i \leq w_i\}.$$

(iii) $\sum_i x_i^* = \bar{\omega} + \sum_j y_j^*$.

The type of equilibrium represented by conditions (i) to (iii) of Proposition 16.G.1 is called a *marginal cost price equilibrium with transfers*. The motivation for this terminology comes from the one-output, one-input case.²⁵

24. That is, the assumptions leading up to conditions (16.F.2) and (16.F.3).

25. We point out that for the general case, the term *marginal cost price equilibrium* is, strictly speaking, inappropriate. Exercise 16.G.3 explains why. However, the terminology is by now standard, and we retain it.

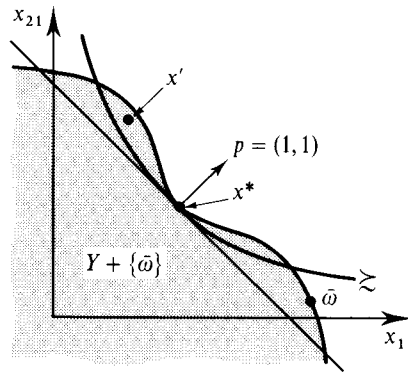


Figure 16.G.2
A marginal cost price equilibrium need not be Pareto optimal.

Exercise 16.G.2: Suppose there are only two goods: an input and an output. Show that in this case, condition (16.G.1) simply says that the price of the input must equal the price of the output multiplied by the marginal productivity of the input or, equivalently, that the price of the output equals its marginal cost.

As we have noted, condition (16.G.1) does *not* imply that the $(y_1^*, \dots, y_J^*)$ are profit-maximizing production plans for price-taking firms. The condition says only that small changes in production plans have no first-order effect on profit. But small changes may still have positive second-order effects (as in Figure 15.C.3, where at a marginal cost price equilibrium the firm actually chooses the production that *minimizes* profits among the efficient productions) and, at any rate, large changes may increase profits (as in Figure 16.G.1). Thus, to achieve allocation (x^*, y^*) may require that a regulatory agency prevent the managers of nonconvex firms from attempting to maximize profits at the given prices p .²⁶

See Quinzii (1992) for extensive background and discussion on the material presented in this section.

It should be noted that the converse result to Proposition 16.G.1, which would assert that every marginal cost price equilibrium is Pareto optimal, is *not true*. In Figure 16.G.2, for example, we show a one-consumer economy with a nonconvex production set. In the figure, x^* is a marginal cost price equilibrium with transfers for the price system $p = (1, 1)$. Yet, allocation x' yields the consumer a higher utility. Informally, this occurs because marginal cost pricing neglects second-order conditions and it may therefore happen that, as at allocation x^* , the second-order conditions for the social utility maximization problem are not fulfilled. As a result, satisfaction of the first-order marginal optimality conditions (which in the case of Figure 16.G.2 amounts simply to the tangency of the indifference curve and the production surface) does not ensure that the allocation is Pareto optimal. (See Exercise 16.G.4 for more on this topic.)

26. In the context of Figure 16.G.1, the regulator could reach the desired outcome by merely prohibiting the firm from shutting down and otherwise letting it maximize profits at the “supporting” prices (assuming that the firm will act as a price taker; otherwise, the regulator may also need to enforce those prices).

APPENDIX A: TECHNICAL PROPERTIES OF THE SET OF FEASIBLE ALLOCATIONS

The set of feasible allocations is

$$A = \{(x_1, \dots, x_I, y_1, \dots, y_J) \in X_1 \times \dots \times X_I \times Y_1 \times \dots \times Y_J : \sum_i x_i = \bar{\omega} + \sum_j y_j\} \\ \subset \mathbb{R}^{LI} \times \mathbb{R}^{LJ}.$$

Our economic problem would not be very interesting if there were no feasible allocations for the economy or if we could give every consumer an unboundedly large consumption vector. We might therefore simply assume that A is nonempty, bounded, and, for good measure, closed (i.e., nonempty and compact). In Chapter 17, where this technical point becomes important for the study of the existence of Walrasian equilibria, we assume exactly this. Nonetheless, it is useful to give, once and for all, a set of sufficient conditions for these very basic properties to hold.

Proposition 16.AA.1: Suppose that

- (i) Every X_i :
 - (i.1) is closed;
 - (i.2) is bounded below (i.e., there is $r > 0$ such that $x_{\ell i} > -r$ for every ℓ and i ; in words, no consumer can supply to the market an arbitrarily large amount of any good).
- (ii) Every Y_j is closed. Moreover, the aggregate production set $Y = \sum_j Y_j$:²⁷
 - (ii.1) is convex;
 - (ii.2) admits the possibility of inaction (i.e., $0 \in Y$);
 - (ii.3) satisfies the no-free-lunch property (i.e., $y \geq 0$ and $y \in Y$ implies $y = 0$);
 - (ii.4) is irreversible ($y \in Y$ and $-y \in Y$ implies $y = 0$).

Then the set of feasible allocations A is closed and bounded [i.e., there is $r > 0$ such that $|x_{\ell i}| < r$ and $|y_{\ell j}| < r$ for all i, j, ℓ and any $(x, y) \in A$]. If, moreover, $-\mathbb{R}_+^L \subset Y$ and we can choose $\hat{x}_i \in X_i$ for every i in such a manner that $\sum_i \hat{x}_i \leq \bar{\omega}$, then A is nonempty.

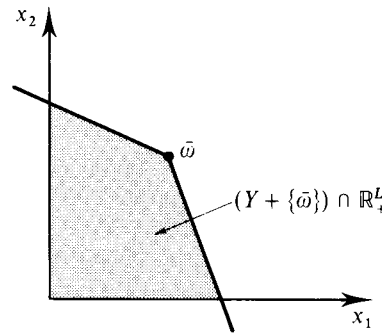
Proof: The proof of this proposition is rather technical, and we shall not give it. Nonetheless, we shall say a few words regarding the logic of the result.

The nonemptiness part is clear enough because we have $\hat{x}_i \in X_i$ for every i and $\sum_i \hat{x}_i - \bar{\omega} \in -\mathbb{R}_+^L \subset Y$. Thus, an allocation with individual consumptions $(\hat{x}_1, \dots, \hat{x}_I)$ and aggregate production vector $\sum_i \hat{x}_i - \bar{\omega}$ is feasible.

Similarly, the closedness of A is a direct consequence of the closedness assumptions on the consumption and production sets (see Exercise 16.AA.1).

What remains is to show that A is bounded. To gain some understanding, suppose that $J = 1$ and $X_i = \mathbb{R}_+^L$ for every i (as long as every X_i is bounded below, the argument for general consumption sets is similar). In Figure 16.AA.1, we represent the set of feasible aggregate consumption bundles $(Y + \{\bar{\omega}\}) \cap \mathbb{R}_+^L$, that is, the set of nonnegative vectors obtained when the origin of Y is shifted to $\bar{\omega} \geq 0$. It is intuitive from the figure that this set can be unbounded above only if Y contains nonnegative, nonzero vectors and so violates the no-free-lunch condition.

27. See Section 5.B for a discussion of conditions (ii.1) to (ii.4).

**Figure 16.AA.1**

The set of feasible aggregate consumption vectors is compact.

You should verify, however, that this intuition also depends on three facts: $0 \in Y$, Y is closed, and Y is convex (see Exercise 16.AA.2). Note now that if the set of feasible aggregate consumptions is bounded, then, a fortiori, so are the sets of feasible individual consumptions and the set of feasible productions (because $J = 1$, we get this set by subtracting $\bar{\omega}$ from each feasible aggregate consumption vector). Hence, A is bounded.

The case with several production sets is more delicate, and it is here that the irreversibility assumption comes to the rescue. Very informally, we can derive, as in the preceding paragraph, the boundedness of feasible *aggregate* productions and feasible individual consumptions. Now, the only way that unboundedness would be possible at the individual production level while remaining bounded in the aggregate is if, so to speak, the unboundedness in one individual production plan was to be canceled by the unboundedness of another. However, this would imply that the collection of all technologies in the economy (i.e., the aggregate production set) allows the reversal of some technologies (see Exercise 16.AA.3 for more details). Incidentally, it can also be shown that irreversibility, with the other assumptions, yields the closedness of Y , so we do not actually need to assume this separately. ■

Proposition 16.AA.2 gives an important implication of the compactness of the set of feasible allocations for the form of the utility possibility set.

Proposition 16.AA.2: Suppose that the set of feasible allocations A is nonempty, closed, and bounded and that utility functions $u_i(\cdot)$ are continuous. Then the utility possibility set U is closed and bounded above.

Proof: Note that $U = U' - \mathbb{R}_+^L$ where

$$U' = \{(u_1(x_1), \dots, u_I(x_I)) : (x, y) \in A\} \subset \mathbb{R}^L.$$

Thus, U' is the image of the compact set A under a continuous function and is therefore itself a compact set (see Section M.F of the Mathematical Appendix). From this, the closedness and the boundedness above of $U = U' - \mathbb{R}_+^L$ follow directly. ■

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EXERCISES

16.C.1^B Show that if a consumption set $X_i \subset \mathbb{R}^L$ is nonempty, closed, and bounded and the preference relation $\succsim_i$ on X_i is continuous, then $\succsim_i$ cannot be locally nonsatiated. [Hint: Show that the continuous utility function representing $\succsim_i$ must have a maximum of X_i ; see Section M.F of the Mathematical Appendix on this.]

16.C.2^A Suppose that the preference relation $\succsim_i$ is locally nonsatiated and that x_i^* is maximal for $\succsim_i$ in set $\{x_i \in X_i: p \cdot x_i \leq w_i\}$. Prove that the following property holds: “If $x_i \succsim_i x_i^*$ then $p \cdot x_i \geq w_i$.”

16.C.3^B In this exercise you are asked to establish the first welfare theorem under a set of assumptions compatible with satiation. Suppose that every X_i is nonempty and convex and that every $\succsim_i$ is strictly convex (i.e., if $x'_i \succsim_i x_i$ and $x'_i \neq x_i$ then $\alpha x'_i + (1 - \alpha)x_i \succ_i x_i$ whenever $0 < \alpha < 1$). Prove the following:

(a) Every i can have at most one global satiation point and preferences are locally nonsatiated at any consumption bundle different from the single global satiation point.

(b) Any price equilibrium with transfers is a Pareto optimum.

16.C.4^A Suppose that for each individual there is a “pleasure function” depending on her own consumption only, given by $u_i(x_i)$. Every individual’s utility depends positively on her own and everyone else’s “pleasure” according to the utility function

$$U_i(x_1, \dots, x_I) = U_i(u_1(x_1), u_2(x_2), \dots, u_I(x_I)).$$

Show that if $x = (x_1, \dots, x_I)$ is Pareto optimal relative to the $U_i(\cdot)$ ’s, then $(x_1, \dots, x_I)$ is also a Pareto optimum relative to the u_i ’s. Does this mean that a community of altruists can use competitive markets to attain Pareto optima? Does your argument depend on the concavity of the u_i ’s, or the U_i ’s?

16.D.1^A Prove that if preferences are locally nonsatiated then the condition: “if $x_i \succ_i x_i^*$ then $p \cdot x_i \geq p \cdot x_i^*$ ” is equivalent to the condition: “ x_i^* is expenditure minimizing for the price vector p in the set $\{x_i \in X_i: x_i \succsim_i x_i^*\}$.”

16.D.2^B Exhibit a one-firm, one-consumer economy in which the production set is convex, the preference relation is continuous and convex, and there is nevertheless a Pareto optimal allocation that can be supported neither as a price equilibrium with transfers nor as a price quasiequilibrium with transfers. Which condition of Proposition 16.D.1 fails?

16.D.3^B Suppose that we have an economy with continuous and strongly monotone preferences (consumption sets are $X_i = \mathbb{R}_+^L$). Suppose also that a strictly positive production is possible; that is, there are $y_j \in Y_j$ such that $\sum_j y_j + \bar{\omega} \gg 0$. Prove that any price quasiequilibrium with transfers must be a price equilibrium with transfers. [Hint: Show first that $w_i > 0$ for some i and then argue that $p \gg 0$.]

16.D.4^C Consider a two-good exchange economy with r identical consumers. The consumption set is $\mathbb{R}_+^2$, the individual endowments are $\omega \in \mathbb{R}_{++}^2$, and the preferences are continuous and strongly monotone but not necessarily convex. Argue that the symmetric allocation in which

every consumer gets her initial allocation is either a Walrasian equilibrium (for some price vector p) or, if it is not, then for r large enough it is not a Pareto optimum. [Hint: the differentiable case is simpler.]

16.E.1^B Given a utility possibility set U , denote by $U' \subset U$ the subset actually achieved by feasible allocations:

$$U' = \{(u_1(x_1), \dots, u_I(x_I)) : \sum_i x_i = \sum_j y_j + \bar{\omega} \text{ for some } y_j \in Y_j\}.$$

(Relative to U' , the set U allows for free disposal of utility).

(a) Give a two-consumer, two-commodity exchange example showing that it is possible for a point of U' to belong to the boundary of U and *not* be a Pareto optimum.

(b) Suppose that every Y_j satisfies free disposal and $0 \in Y_j$. Also, assume that for every i , $X_i = \mathbb{R}_+^I$ and $\succeq_i$ is continuous and strongly monotone. Show that any boundary point of U that belongs to U' is then a Pareto optimum. [Hint: Let $u_i(0) = 0$ for all i and show first that $U' = U \cap \mathbb{R}_+^L$. Next argue that if $u \in U$ is a Pareto optimum and $0 \leq u' \leq u$, $u' \neq u$, then we must be able to reach u' with a surplus of goods relative to u .]

(c) Consider an exchange economy with consumption sets equal to $\mathbb{R}_+^L$, continuous, locally nonsatiated preferences, and a strictly positive total endowment vector. Show that if $u = (u_1, \dots, u_I)$ is the utility vector corresponding to a price quasiequilibrium with transfers then u cannot be in the interior of U ; that is, there is no feasible allocation yielding higher utility to *every* consumer. [Hint: Show that $w_i > 0$ for some i and then apply Proposition 16.D.2.]

16.E.2^B Show that the utility possibility set U of an economy with convex production and consumption sets and with concave utility functions is convex.

16.F.1^B In text.

16.F.2^A Derive the first-order conditions (16.F.2) and (16.F.3) of the maximization problem (16.F.1).

16.F.3^A Derive conditions (16.F.4), (16.F.5), and (16.F.6) from the first-order conditions (16.F.2) and (16.F.3).

16.F.4^A Derive the first-order conditions (16.F.4), (16.F.5), and (16.F.6) from problems (16.F.7), (16.F.8), and (16.F.9), respectively.

16.G.1^A Prove Proposition 16.G.1 using the first-order conditions (16.F.2) and (16.F.3).

16.G.2^A In text.

16.G.3^B Exhibit graphically a one-consumer, one-firm economy with two inputs and one output where at the (unique) marginal cost price equilibrium, cost is *not* minimized. [Hint: Choose the production function to violate quasiconcavity.]

16.G.4^B Show that under the general conditions of Section 16.G if there is a single consumer (perhaps a normative representative consumer) with convex preferences, then there exists at least one marginal cost price equilibrium that is an optimum.

16.G.5^B In a certain economy there are two commodities, education (e) and food (f), produced by using labor (L) and land (T) according to the production functions

$$e = (\text{Min} \{L, T\})^2 \quad \text{and} \quad f = (LT)^{1/2}$$

There is a single consumer with the utility function

$$u(e, f) = e^\alpha f^{(1-\alpha)},$$

and endowment (ω_L, ω_T) . To ease the calculations, take $\omega_L = \omega_T = 1$ and $\alpha = \frac{1}{2}$.

(a) Find the optimal allocation of the endowments to their productive uses.

(b) Recognizing that the production of education entails increasing returns to scale, the government of this economy decides to control the education industry and finance its operation with a lump-sum tax on the consumer. The consumption of education is competitive in the sense that the consumer can choose any amount of education and food desired at the going prices. The food industry remains competitive in both its production and consumption aspects. Assuming that the education industry minimizes cost, find the marginal cost of education at the optimum. Show that if this price of education were announced, together with a lump-sum tax to finance the deficit incurred when the education sector produces the optimal amount at this price for its product, then the consumer's choice of education will be at the optimal level.

(c) What is the level of the lump-sum tax necessary to decentralize this optimum in the manner described in (b)?

Now suppose there are two consumers and that their preferences are identical to those above. One owns all of the land and the other owns all of the labor. In this society, arbitrary lump-sum taxes are not possible. It is the law that any deficit incurred by a public enterprise must be covered by a tax on the value of land.

(d) In appropriate notation, write the transfer from the landowner as a function of the government's planned production of education.

(e) Find a marginal cost price equilibrium for this economy where transfers have to be compatible with the transfer function specified in (d). Is it Pareto optimal?

16.AA.1^A Show that if every X_i and every Y_j is closed, then the set A of feasible allocations is closed.

16.AA.2^B Show that $(Y + \{\bar{\omega}\}) \cap \mathbb{R}_+^L$ is compact if the following four assumptions are satisfied: (i) Y is closed, (ii) Y is convex, (iii) $0 \in Y$, and (iv) if $v \in Y \cap \mathbb{R}_+^L$ then $v = 0$. Exhibit graphically four examples showing that each of the four assumptions is indispensable.

16.AA.3^B Suppose that $Y = Y_1 + Y_2 \subset \mathbb{R}_+^L$ satisfies the assumptions given in Exercise 16.AA.2 and that $0 \in Y_1$, $0 \in Y_2$. Argue that if the irreversibility assumption holds for Y then $\{y_1 \in Y_1: y_1 + y_2 + \bar{\omega} \geq 0 \text{ for some } y_2 \in Y_2\}$ is bounded.

The Positive Theory of Equilibrium

17.A Introduction

In this chapter, we study the theoretical predictive power of the Walrasian equilibrium model. Thus, in contrast with Chapter 16, our outlook here is positive rather than normative.

We begin in Section 17.B by laying the foundations for our analysis. We recall the basic model of a *private ownership economy* and the definition of a *Walrasian equilibrium* presented in Section 16.B. We then introduce the notion of an *aggregate excess demand function* and, in a framework of strong assumptions, we characterize Walrasian equilibria as solutions to a system of aggregate excess demand equations. The analysis of this system of equations serves throughout the chapter as our primary method for the study of Walrasian equilibria. (In Appendix A, we discuss another useful equation system based instead on the welfare properties of Walrasian equilibria.)

In Section 17.C (and with much more generality in Appendix B), we present conditions guaranteeing the existence of a Walrasian equilibrium. The identification of an interesting set of conditions ensuring existence assures us that the object of our study in this chapter is not vacuous. A key condition turns out to be the *convexity* of the decision problems of individual economic agents.

Sections 17.D to 17.H all deal with properties of the set of equilibria. Section 17.D reaches a general conclusion: typically (or in the usual terminology, *generically*), there is a finite number of equilibria, and each equilibrium is therefore “locally isolated.” Even more, this number is odd, and the equilibria fall naturally into two categories according to the sign of their “index.” Section 17.E brings bad news: without further assumptions on the nature of preferences, endowments, or technologies, we cannot say anything more; the behavior of excess demand functions and hence the properties of Walrasian equilibria are not restricted in any manner that goes beyond the facts established in Section 17.D. This negative message reverberates in Section 17.F on uniqueness, Section 17.G on comparative statics, and Section 17.H on (*tâtonnement*) stability. The purpose of these three sections is precisely to find interesting sufficient conditions for, respectively, the uniqueness of equilibria, good comparative statics properties, and the stability of equilibria. A common theme of

the three sections is the role of two sufficient conditions: the *weak axiom of revealed preference in the aggregate* (a way of saying that wealth effects do not cancel in the aggregate the positive influence of the substitution effects), and the property of *gross substitution* (a way of saying that there are not strong complementarities among the goods in the economy).

In Section 17.I, we return to the role of convexity in guaranteeing the existence of Walrasian equilibrium. We qualify this role by showing that nonconvexities that are “small” relative to the aggregate economy (e.g., the indivisibility represented by a car) are not an obstacle to the (near) existence of equilibria, even if they are “large” from the standpoint of an individual agent.

This chapter is of interest from both methodological and substantive points of view. From a substantive standpoint, it deals with an important theory: that of Walrasian equilibrium. Methodologically, the questions that we ask (e.g., does an equilibrium exist? Are the equilibria typically isolated? Is the equilibrium unique? Is it stable? What are the effects of shocks?) and the techniques that we use are questions and techniques that are of relevance to any theory of equilibrium.

17.B Equilibrium: Definitions and Basic Equations

The concept of a private ownership economy was described in Section 16.B. In such an economy, there are I consumers and J firms. Every consumer i is specified by a consumption set $X_i \subset \mathbb{R}^L$, a preference relation $\succeq_i$ on X_i , an initial endowment vector $\omega_i \in \mathbb{R}^L$, and an ownership share $\theta_{ij} \geq 0$ of each firm $j = 1, \dots, J$ (where $\sum_i \theta_{ij} = 1$). Each firm j is characterized by a production set $Y_j \subset \mathbb{R}^L$. An *allocation* for such an economy is a collection of consumption and production vectors:

$$(x, y) = (x_1, \dots, x_I, y_1, \dots, y_J) \in X_1 \times \dots \times X_I \times Y_1 \times \dots \times Y_J.$$

The object of investigation in this chapter is the notion of *Walrasian equilibrium*, which we take as a positive prediction for the outcome of a system of markets in which consumers and firms are price takers and the wealth of consumers derives from their initial endowments and profit shares. The formal notion of a Walrasian equilibrium was already introduced in Definition 16.B.3. Definition 17.B.1 repeats it.

Definition 17.B.1: Given a private ownership economy specified by

$$(\{(X_i, \succeq_i)\}_{i=1}^I, \{Y_j\}_{j=1}^J, \{(\omega_i, \theta_{i1}, \dots, \theta_{iJ})\}_{i=1}^I),$$

an allocation (x^*, y^*) and a price vector $p = (p_1, \dots, p_L)$ constitute a *Walrasian* (or *competitive*, or *market*, or *price-taking*) equilibrium if

(i) For every j , $y_j^* \in Y_j$ maximizes profits in Y_j ; that is,

$$p \cdot y_j \leq p \cdot y_j^* \quad \text{for all } y_j \in Y_j.$$

(ii) For every i , $x_i^* \in X_i$ is maximal for $\succeq_i$ in the budget set

$$\{x_i \in X_i: p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*\}^1$$

(iii) $\sum_i x_i^* = \sum_i \omega_i + \sum_j y_j^*$.

1. The terminology “ x_i is maximal for $\succeq_i$ in set B ” means that x_i is consumer i ’s preference-maximizing choice in the set B ; that is, $x_i \in B$ and $x_i \succeq_i x_i'$ for all $x_i' \in B$.

For purposes of formal analysis, it is extremely helpful to be able to express equilibria as the solutions of a system of equations. We devote the remainder of this section to the study of how this may be done. In what follows, we aim at being very concrete and impose strong assumptions to simplify the analysis.

Exchange Economies and Excess Demand Functions

We begin our derivation of equilibrium equations by studying the case of a pure exchange economy. Recall that a pure exchange economy is one in which the only possible production activities are those of free disposal. Formally, we let $J = 1$ and $Y_1 = -\mathbb{R}_+^L$. We take $X_i = \mathbb{R}_+^L$ and we assume at the outset that each consumer's preferences are *continuous*, *strictly convex*, and *locally nonsatiated* (shortly we shall strengthen local nonsatiation to strong monotonicity). We also assume that $\sum_i \omega_i \gg 0$.

For a pure exchange economy satisfying the above assumptions, the three conditions of Definition 17.B.1 can be equivalently restated as: $(x^*, y^*) = (x_1^*, \dots, x_I^*, y_1^*)$ and $p \in \mathbb{R}^L$ constitute a Walrasian equilibrium if and only if

- (i') $y_1^* \leq 0$, $p \cdot y_1^* = 0$, and $p \geq 0$.
- (ii') $x_i^* = x_i(p, p \cdot \omega_i)$ for all i [where $x_i(\cdot)$ is consumer i 's Walrasian demand function].
- (iii') $\sum_i x_i^* - \sum_i \omega_i^* = y_1^*$.

Condition (i') is the only one that is not immediate. In Exercise 17.B.1, you are asked to show that it is equivalent to condition (i) of Definition 17.B.1.

Conditions (i') to (iii') yield the simple result of Proposition 17.B.1.

Proposition 17.B.1: In a pure exchange economy in which consumer preferences are continuous, strictly convex, and locally nonsatiated, $p \geq 0$ is a Walrasian equilibrium price vector if and only if:

$$\sum_i (x_i(p, p \cdot \omega_i) - \omega_i) \leq 0. \quad (17.B.1)$$

Proof: That (17.B.1) must hold in any Walrasian equilibrium of such an economy follows immediately from conditions (i') to (iii'). In the other direction, suppose that (17.B.1) holds. If we let $y_1^* = \sum_i (x_i(p, p \cdot \omega_i) - \omega_i)$ and $x_i^* = x_i(p, p \cdot \omega_i)$, then $(x_1^*, \dots, x_I^*, y_1^*)$ and p satisfy conditions (i') to (iii'). In particular, note that $p \cdot y_1^* = p \cdot \sum_i (x_i(p, p \cdot \omega_i) - \omega_i) = \sum_i (p \cdot x_i(p, p \cdot \omega_i) - p \cdot \omega_i) = 0$, where the last equality follows because with local nonsatiation we have $p \cdot x_i(p, p \cdot \omega_i) = p \cdot \omega_i$ for all i . ■

The vector $x_i(p, p \cdot \omega_i) - \omega_i \in \mathbb{R}^L$ lists consumer i 's net, or *excess*, demand for each good over and above the amount that he possesses in his endowment vector ω_i . Condition (17.B.1) suggests that it may be useful to have a formal representation of this excess demand vector, and of its sum over the I consumers, as a function of prices. This is given in Definition 17.B.2.

Definition 17.B.2: The *excess demand function of consumer i* is

$$z_i(p) = x_i(p, p \cdot \omega_i) - \omega_i,$$

where $x_i(p, p \cdot \omega_i)$ is consumer i 's Walrasian demand function. The (*aggregate*)

excess demand function of the economy is

$$z(p) = \sum_i z_i(p).$$

The domain of this function is a set of nonnegative price vectors that includes all strictly positive price vectors.

Using the economy's excess demand function $z(p)$, condition (17.B.1) can now be expressed more succinctly as follows:

$$“p \in \mathbb{R}_+^L \text{ is an equilibrium price vector if and only if } z(p) \leq 0.” \quad (17.B.1')$$

Note that if p is a Walrasian equilibrium price vector in a pure exchange economy with locally nonsatiated preferences, then $p \geq 0$, $z(p) \leq 0$, and $p \cdot z(p) = \sum_i p \cdot z_i(p) = \sum_i (p \cdot x_i(p, p \cdot \omega_i) - p \cdot \omega_i) = 0$ (the last equality follows once again from local nonsatiation). Therefore, for every ℓ , we not only have $z_\ell(p) \leq 0$, but also $z_\ell(p) = 0$ if $p_\ell > 0$. Thus, we see that at an equilibrium a good ℓ can be in excess supply (i.e., have $z_\ell(p) < 0$), but only if it is free (i.e., only if $p_\ell = 0$).²

To simplify matters even more, we go one step further by assuming that consumer preferences are *strongly monotone*. Thus, for the rest of this section (and, in fact, for all sections of this chapter except Section 17.I and Appendix B), we let $X_i = \mathbb{R}_+^L$ for all i and assume that *all preference relations $\succsim_i$ are continuous, strictly convex, and strongly monotone*.

With strongly monotone preferences, any Walrasian equilibrium must involve a *strictly positive* price vector $p \gg 0$; otherwise consumers would demand an unboundedly large quantity of all the free goods. As a result, we conclude that with strong monotonicity of preferences, a price vector $p = (p_1, \dots, p_L)$ is a *Walrasian equilibrium price vector* if and only if it “clears all markets”; that is, if and only if it solves the system of L equations in L unknowns:

$$z_\ell(p) = 0 \quad \text{for every } \ell = 1, \dots, L, \quad (17.B.2)$$

or, in more compact notation, $z(p) = 0$.

Throughout this chapter, we study the properties of Walrasian equilibria largely by examining the properties of the system of equilibrium equations (17.B.2). We should point out, however, that this is not the only system of equations that we could use to characterize Walrasian equilibria. In Appendix A, for example, we discuss an important alternative system that exploits the welfare properties of Walrasian equilibria identified in Chapter 16.

Proposition 17.B.2 enumerates the properties of the aggregate excess demand function, in pure exchange economies with strongly monotone preferences, that are essential to the developments of this chapter.

Proposition 17.B.2: Suppose that, for every consumer i , $X_i = \mathbb{R}_+^L$ and $\succsim_i$ is continuous, strictly convex, and strongly monotone. Suppose also that $\sum_i \omega_i \gg 0$. Then the aggregate excess demand function $z(p)$, defined for all price vectors $p \gg 0$,

2. As a simple example, good ℓ might be a “bad.” Then, we would expect that good ℓ 's price would be zero, consumer demand for the good would be zero, and the excess supply $z_\ell(p) = \omega_\ell > 0$ would be eliminated using the disposal technology.

satisfies the properties:

- (i) $z(\cdot)$ is continuous.
- (ii) $z(\cdot)$ is homogeneous of degree zero.
- (iii) $p \cdot z(p) = 0$ for all p (Walras' law).
- (iv) There is an $s > 0$ such that $z_\ell(p) > -s$ for every commodity ℓ and all p .
- (v) If $p^n \rightarrow p$, where $p \neq 0$ and $p_\ell = 0$ for some ℓ , then

$$\text{Max} \{z_1(p^n), \dots, z_L(p^n)\} \rightarrow \infty.$$

Proof: With the exception of property (v), all these properties are direct consequences of the definition and the parallel properties of demand functions.³ The bound in (iv) follows from the nonnegativity of demand (i.e., the fact that $X_i = \mathbb{R}_+^L$), which implies that a consumer's total net supply to the market of any good ℓ can be no greater than his initial endowment. You are asked to prove property (v) in Exercise 17.B.2. The intuition for it is this: As some prices go to zero, a consumer whose wealth tends to a strictly positive limit [note that, because $p \cdot (\sum_i \omega_i) > 0$, there must be at least one such consumer] and with strongly monotone preferences will demand an increasingly large amount of some of the commodities whose prices go to zero (but perhaps not of all such commodities: relative prices still matter). ■

Finally, note that because of Walras' law, to verify that a price vector $p \gg 0$ clears all markets [i.e., has $z_\ell(p) = 0$ for all ℓ] it suffices to check that it clears *all markets but one*. In particular, if $p \gg 0$ and $z_1(p) = \dots = z_{L-1}(p) = 0$, then because $p \cdot z(p) = \sum_\ell p_\ell z_\ell(p) = 0$ and $p_L > 0$, we must also have $z_L(p) = 0$. Hence, if we denote the vector of $L - 1$ excess demands for goods 1 through $L - 1$ by

$$\hat{z}(p) = (z_1(p), \dots, z_{L-1}(p)),$$

we see that a strictly positive price vector p is a Walrasian equilibrium if and only if $\hat{z}(p) = 0$.

Production Economies

It is possible to extend the methodology based on excess demand equations to the general production case. Assume, to begin with, that production sets are closed, strictly convex, and bounded above. Then, for any price vector $p \gg 0$, we can let $\pi_j(p)$ and $y_j(p)$ be the maximum profits and the profit-maximizing production vector, respectively, for firm j . Defining

$$\bar{z}(p) = \sum_i x_i(p, p \cdot \omega_i + \sum_j \theta_{ij} \pi_j(p)) - \sum_i \omega_i - \sum_j y_j(p) \quad (17.B.3)$$

as the *production inclusive excess demand function*, we see that p is a Walrasian equilibrium price vector if and only if it solves the system of equations $\bar{z}(p) = 0$. In Exercise 17.B.4, you are asked to show that under a weak hypothesis (that a strictly positive aggregate consumption bundle is producible using the initial endowments), the function $\bar{z}(\cdot)$ satisfies properties (i) to (v) of Proposition 17.B.2.

Note that if the production sets are not bounded above, then $\bar{z}(p)$ may fail to be

3. Note, incidentally, that properties (i) to (iv) continue to hold even if preferences are only locally nonsatiated.

defined for some $p \gg 0$ [because we may have $\pi_j(p) = \infty$ for some j]. Nevertheless, an equilibrium price vector is still characterized by $\bar{z}(p) = 0$.

When production sets are not strictly convex, matters become more complicated because the correspondences $y_j(p)$ may no longer be single-valued. Indeed, a production situation of considerable theoretical and practical importance—and one that we certainly do not want to rule out by assumption—is the case of constant returns to scale. With constant returns, however, production sets are neither strictly convex nor bounded above (except for the trivial case in which no positive amount of any good can be produced). In principle, we could still view the equilibria as the zeros of a “production inclusive excess demand correspondence,” defined as in (17.B.3) for a subset of strictly positive prices.⁴ Correspondences, however, do not make good equational systems (e.g., they cannot be differentiated). It is therefore usually much more convenient in such cases to capture the equilibria as the solutions of an extended system of equations involving the production and the consumption sides of the economy. We illustrate this idea in the small type discussion that follows.

To see how an extended system of equations can be constructed, consider the case in which production is of the linear activity type (this case is reviewed in Appendix A of Chapter 5). Say that, in addition to the disposal technologies, we have J basic activities $a_1, \dots, a_J \in \mathbb{R}^L$. That is, the aggregate production set is

$$Y = \left\{ y \in \mathbb{R}^L : y \leq \sum_j \alpha_j a_j \text{ for some } (\alpha_1, \dots, \alpha_J) \geq 0 \right\}.$$

Because preferences are strongly monotone, there can be no free goods at an equilibrium (i.e., we must have $p \gg 0$). Also, productions should be profit maximizing, and because of constant returns, these maximum profits must be zero. Therefore, a pair (p, α) formed by a price vector $p \in \mathbb{R}_+^L$ and a vector of activity levels $\alpha \in \mathbb{R}_+^J$ constitute an equilibrium if and only if they solve

$$z(p) - \sum_j \alpha_j a_j = 0 \quad (17.B.4)$$

and

$$p \cdot a_j \leq 0, \quad \alpha_j (p \cdot a_j) = 0 \quad \text{for all } j, \quad (17.B.5)$$

where $z(\cdot)$ is the consumers' aggregate excess demand function of Definition 17.B.2. Note that, if so desired, condition (17.B.5) can be expressed as a system of equations: just replace “ $p \cdot a_j \leq 0, \alpha_j (p \cdot a_j) = 0$ ” by “ $\alpha_j p \cdot a_j + \text{Max}\{0, p \cdot a_j\} = 0$.” Exercise 17.B.5 presents an extension of the current discussion to a more general production case allowing for continuous substitution of activities.

It is worth emphasizing that, at least for the case of convex technologies, there would not be much loss of conceptual generality if we assumed that the production sector of the economy was composed of a single firm endowed with a constant returns production technology. To see this, recall from Proposition 5.B.2 that by creating for each firm j an extra, firm-specific, factor of production, we can always assume that every Y_j exhibits constant returns (when we do so, we transform each consumer's ownership shares of the profits of the j th firm into endowments of the j th new physical resource). Because profits at an equilibrium must then be zero, we see that once this is done there is no need to keep the identity of firms separate in order to compute the wealth of consumers. Moreover, from the point of view of production decisions, again no such need arises. As we saw in Section 5.E, we could as well work with the aggregate “representative firm” $Y = \sum_j Y_j$.

4. That is, p would be an equilibrium price vector if and only if $0 \in \bar{z}(p)$.

A single constant returns Y can be interpreted as a description of a long-run state of knowledge that is freely available to every agent in the economy (i.e., to every consumer), for the purpose of setting up firms or simply for household production. In fact, we could go one step further and formally dispense with the separate consideration of firms and of the profit-maximization condition. In Exercise 17.B.6, you are asked to show that a Walrasian equilibrium can be redefined in terms of the following two-stage process: first, consumers choose a vector $v_i \in \mathbb{R}^L$ subject to the budget constraint $p \cdot v_i \leq p \cdot \omega_i$ (the equilibrium market-clearing condition is $\sum_i v_i = \sum_i \omega_i$); second, every i uses v_i and the technology Y for household production of a most preferred consumption bundle.⁵

17.C Existence of Walrasian Equilibrium

When studying a positive theory, the first question to ask is: under what conditions does the formal model possess a solution? That is, is it capable of predicting a definite outcome? This is known as the *existence* problem. Conceptually, the assurance of existence of an equilibrium means that our equilibrium notion passes the logical test of consistency. It tells us that the mathematical model is well suited to the purposes it has been designed for. Although an existence theorem can hardly be the end of the story, it is, in a sense, the door that opens into the house of analysis.⁶

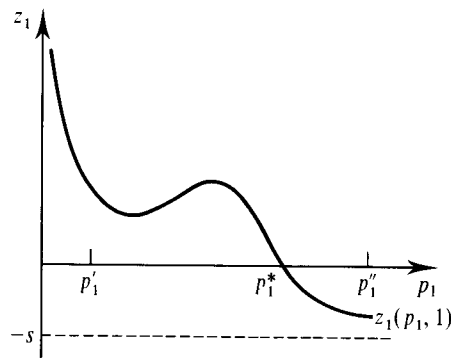
The existence of a Walrasian equilibrium can be established in considerable generality. To maintain the natural flow of exposition, in this section we offer a detailed examination of the existence problem for the particular case that will be our primary focus throughout the chapter: pure exchange economies modeled by means of excess demand functions. In Appendix B, we discuss the existence problem in the general case.

We have seen in Section 17.B that the excess demand function $z(\cdot)$ of an exchange economy with $\sum_i \omega_i \gg 0$ and continuous, strictly convex, and strongly monotone preferences satisfies properties (i) to (v) of Proposition 17.B.2. We now argue that *any* function $z(\cdot)$ satisfying these five conditions admits a solution, that is, a price vector p such that $z(p) = 0$. By doing so, we establish that a Walrasian equilibrium exists under the conditions of Proposition 17.B.2.

To start simply, suppose that there are only two commodities (i.e., $L = 2$). For this case, it is an easy matter to establish the existence of an equilibrium. First, by the homogeneity of degree zero of $z(\cdot)$ (condition (ii) of Proposition 17.B.2), we can normalize $p_2 = 1$ and look for equilibrium price vectors of the form $(p_1, 1)$. Then, by Walras' law (condition (iii) of Proposition 17.B.2), an equilibrium can be obtained as a solution to the single equation $z_1(p_1, 1) = 0$. This one-variable problem is

5. This process formally reduces the production economy to an economy where only exchange takes place. But we do not mean to suggest that the induced exchange economy satisfies all of the strong assumptions that we have imposed in this chapter.

6. It should be emphasized that finding a class of conditions that guarantee the existence of a Walrasian equilibrium does not say that this is the outcome that will occur whenever preferences, endowments, and technologies satisfy the assumptions of the existence theorem: the behavioral assumption of price taking and the institutional assumptions of complete markets must also hold. However, when the conditions required for existence are *not* satisfied by preferences, endowments, or technologies, it does suggest that the type of equilibrium under consideration may not be the right one to look for.

**Figure 17.C.1**

Proof of existence of an equilibrium for the case $L = 2$.

represented in Figure 17.C.1.⁷ When p_1' is very small, we must have $z_1(p_1', 1) > 0$; if p_1'' is very large, we have $z_1(p_1'', 1) < 0$. These two boundary restrictions follow by using conditions (iv) and (v) of Proposition 17.B.2 to identify the commodity with positive excess demand as the one whose relative price is very low.⁸ Because the function $z_1(p_1, 1)$ is continuous [condition (i) of Proposition 17.B.2], there must be an intermediate value $p_1^* \in [p_1', p_1'']$ with $z_1(p_1^*, 1) = 0$ and, hence, an equilibrium price vector must exist.

In the general case of more than two commodities, the proof that a solution exists is more delicate, and involves the use of some powerful mathematical tools. In Proposition 17.C.1, we follow a traditional approach that invokes Kakutani's fixed-point theorem (see Section M.I of the Mathematical Appendix). We should point out that the proof of Proposition 17.C.1 has to deal with the technical complication that excess demand is not defined when the prices of some commodities are zero. The reader may actually gain a more direct insight into the nature of the fixed-point argument from Proposition 17.C.2, which contains a very simple proof for the case of excess demand functions defined for all nonzero, nonnegative prices.

Proposition 17.C.1: Suppose that $z(p)$ is a function defined for all strictly positive price vectors $p \in \mathbb{R}_{++}^L$ and satisfying conditions (i) to (v) of Proposition 17.B.2. Then the system of equations $z(p) = 0$ has a solution. Hence, a Walrasian equilibrium exists in any pure exchange economy in which $\sum_i \omega_i \gg 0$ and every consumer has continuous, strictly convex, and strongly monotone preferences.

7. Note that we revert to the usual mathematical convention of representing the independent variable p_1 on the horizontal axis. The partial equilibrium convention of putting prices on the vertical axis, to which we have adhered throughout Part III, is a vestige of the origins of the theory in Marshall (1920) where, in contrast with Walras (1874), prices are, in fact, dependent variables.

8. In particular, property (iv) implies that the value of intended sales is bounded. By Walras' law, the value of intended purchases must therefore also be bounded. Because, by property (v), intended purchases become unbounded in physical terms for *some* good as $p_1 \rightarrow 0$, it follows that it must be good 1 whose demand becomes unbounded as $p_1 \rightarrow 0$. Hence, $z_1(p_1, 1) > 0$ for p_1' sufficiently small. By symmetry, as $p_1 \rightarrow \infty$ [which, by the homogeneity of degree zero of $z(\cdot)$, is equivalent to $p_2 \rightarrow 0$ holding p_1 fixed], for p_1'' large enough we must have $z_2(p_1'', 1) > 0$, and therefore $z_1(p_1'', 1) < 0$.

Proof: We begin by normalizing prices in a convenient way. Denote by

$$\Delta = \left\{ p \in \mathbb{R}_+^L : \sum_{\ell} p_{\ell} = 1 \right\}$$

the unit simplex in $\mathbb{R}^L$. Because the function $z(\cdot)$ is homogeneous of degree zero, we can restrict ourselves, in our search for an equilibrium, to price vectors in Δ . Note, however, that the function $z(\cdot)$ is well defined only for price vectors in the set

$$\text{Interior } \Delta = \{ p \in \Delta : p_{\ell} > 0 \text{ for all } \ell \}.$$

We shall proceed in five steps. In the first two, we construct a certain correspondence $f(\cdot)$ from Δ to Δ . In the third, we argue that any fixed point of $f(\cdot)$, that is, any p^* with $p^* \in f(p^*)$, has $z(p^*) = 0$. The fourth step proves that $f(\cdot)$ is convex valued and upper hemicontinuous (or, equivalently, that it has a closed graph). Finally, the fifth step applies Kakutani's fixed-point theorem to show that a p^* with $p^* \in f(p^*)$ necessarily exists.

For notational clarity, in defining the set $f(p) \subset \Delta$, we denote the vectors that are elements of $f(p)$ by the symbol q .

Step 1: Construction of the fixed-point correspondence for $p \in \text{Interior } \Delta$. Whenever $p \gg 0$, we let

$$f(p) = \{ q \in \Delta : z(p) \cdot q \geq z(p) \cdot q' \text{ for all } q' \in \Delta \}.$$

In words: Given the current "proposal" p , the "counterproposal" assigned by the correspondence $f(\cdot)$ is any price vector q that, among the permissible price vectors (i.e., among the members of Δ), maximizes the value of the excess demand vector $z(p)$. This makes economic sense; thinking of $f(\cdot)$ as a rule that adjusts current prices in a direction that eliminates any excess demand, the correspondence $f(\cdot)$ as defined above assigns the highest prices to the commodities that are most in excess demand. In particular, we have

$$f(p) = \{ q \in \Delta : q_{\ell} = 0 \text{ if } z_{\ell}(p) < \text{Max} \{ z_1(p), \dots, z_L(p) \} \}.$$

Observe that if $z(p) \neq 0$ for $p \gg 0$, then because of Walras' law we have $z_{\ell}(p) < 0$ for some ℓ and $z_{\ell'}(p) > 0$ for some $\ell' \neq \ell$. Thus, for such a p , any $q \in f(p)$ has $q_{\ell} = 0$ for some ℓ . Therefore, if $z(p) \neq 0$ then $f(p) \subset \text{Boundary } \Delta = \Delta \setminus \text{Interior } \Delta$. In contrast, if $z(p) = 0$ then $f(p) = \Delta$.

Step 2: Construction of the fixed-point correspondence for $p \in \text{Boundary } \Delta$. If $p \in \text{Boundary } \Delta$, we let

$$f(p) = \{ q \in \Delta : p \cdot q = 0 \} = \{ q \in \Delta : q_{\ell} = 0 \text{ if } p_{\ell} > 0 \}.$$

Because $p_{\ell} = 0$ for some ℓ , we have $f(p) \neq \emptyset$. Note also that with this construction, no price from Boundary Δ can be a fixed point; that is, $p \in \text{Boundary } \Delta$ and $p \in f(p)$ cannot occur because $p \cdot p > 0$ while $p \cdot q = 0$ for all $q \in f(p)$.

Step 3: A fixed point of $f(\cdot)$ is an equilibrium. Suppose that $p^* \in f(p^*)$. As we pointed out in step 2, we must have $p^* \notin \text{Boundary } \Delta$. Therefore $p^* \gg 0$. If $z(p^*) \neq 0$, then we saw in step 1 that $f(p^*) \subset \text{Boundary } \Delta$, which is incompatible with $p^* \in f(p^*)$ and $p^* \gg 0$. Hence, if $p^* \in f(p^*)$ we must have $z(p^*) = 0$.

Step 4: The fixed-point correspondence is convex-valued and upper hemicontinuous. To establish convex-valuedness, note that, both when $p \in \text{Interior } \Delta$ and when $p \in \text{Boundary } \Delta$, $f(p)$ equals a level set of a linear function defined on the convex set Δ [that is, a set of the form $\{q \in \Delta: \lambda \cdot q = k\}$ for some scalar k and vector $\lambda \in \mathbb{R}^L$], and so it is convex. (Exercise 17.C.1 asks for a more explicit verification.)⁹

To establish upper hemicontinuity (see Section M.M of the Mathematical Appendix for definitions), consider sequences $p^n \rightarrow p$, $q^n \rightarrow q$ with $q^n \in f(p^n)$ for all n . We have to show that $q \in f(p)$. There are two cases: $p \in \text{Interior } \Delta$ and $p \in \text{Boundary } \Delta$.

If $p \in \text{Interior } \Delta$, then $p^n \gg 0$ for n sufficiently large. From $q^n \cdot z(p^n) \geq q' \cdot z(p^n)$ for all $q' \in \Delta$ and the continuity of $z(\cdot)$, we get $q \cdot z(p) \geq q' \cdot z(p)$ for all q' ; that is, $q \in f(p)$.

Now suppose that $p \in \text{Boundary } \Delta$. Take any ℓ with $p_\ell > 0$. We shall argue that for n sufficiently large we have $q_\ell^n = 0$ and therefore it must be that $q_\ell = 0$; from this, $q \in f(p)$ follows. Because $p_\ell > 0$, there is an $\varepsilon > 0$ such that $p_\ell^n > \varepsilon$ for n sufficiently large. If, in addition, $p^n \in \text{Boundary } \Delta$ then $q_\ell^n = 0$ by the definition of $f(p^n)$. If, instead, $p^n \gg 0$ then the boundary conditions (iv) and (v) of Proposition 17.B.2 come into play. They imply that, for n sufficiently large, we must have

$$z_\ell(p^n) < \text{Max} \{z_1(p^n), \dots, z_L(p^n)\}$$

and therefore that, again, $q_\ell^n = 0$. To prove the above inequality, note that by condition (v) the right-hand side of the above expression goes to infinity with n (because $p \in \text{Boundary } \Delta$, some prices go to zero as $n \rightarrow \infty$). But the left-hand side is bounded above because if it is positive then

$$z_\ell(p^n) \leq \frac{1}{\varepsilon} p_\ell^n z_\ell(p^n) = -\frac{1}{\varepsilon} \sum_{\ell' \neq \ell} p_{\ell'}^n z_{\ell'}(p^n) < \frac{s}{\varepsilon} \sum_{\ell' \neq \ell} p_{\ell'}^n < \frac{s}{\varepsilon},$$

where s is the bound in excess supply given by condition (iv).¹⁰ In summary, for p^n close enough to Boundary Δ , the maximal demand corresponds to some of the commodities whose price is close to zero. Therefore, we conclude that, for large n , any $q^n \in f(p^n)$ will put nonzero weight only on commodities whose prices approach zero. But this guarantees $p \cdot q = 0$, and so $q \in f(p)$.

Step 5: A fixed point exists. Kakutani's fixed-point theorem (see Section M.I of the Mathematical Appendix) says that a convex-valued, upper hemicontinuous correspondence from a nonempty, compact, convex set into itself has a fixed point. Since Δ is a nonempty, convex, and compact set, and since $f(\cdot)$ is a convex-valued upper hemicontinuous correspondence from Δ to Δ , we conclude that there is a $p^* \in \Delta$ with $p^* \in f(p^*)$. ■

It is instructive to examine which of properties (i) to (v) of Proposition 17.B.2 fail to hold for the excess demand functions corresponding to the Edgeworth boxes of Figures 15.B.10(a) and (b), where, as we saw, no Walrasian equilibrium existed. In

9. Note also that for any $p \in \Delta$, the set $f(p)$ is always a face of the simplex Δ ; that is, it is one of the subsets of Δ spanned by a finite subset of unit coordinates. For $p \in \text{Boundary } \Delta$, $f(p)$ is the face of Δ spanned by the zero coordinates of p . For $p \in \text{Interior } \Delta$, $f(p)$ is the face spanned by the coordinates corresponding to commodities with maximal excess demand.

10. In words, the last chain of inequalities says that the expenditure on commodity ℓ is bounded because it has to be financed by, and therefore cannot be larger than, the bounded value of excess supplies.

the case of Figure 15.B.10(b), preferences are not convex and therefore $z(\cdot)$ is not a function, let alone a continuous one (condition (i)).¹¹ For Figure 15.B.10(a), it is property (v) that fails: For any sequence of prices $(p_1^n, p_2^n) \rightarrow (1, 0)$, excess demand stays bounded. Note that in the limit there is a single consumer with positive wealth, but the preferences for this consumer, while monotone, are not strongly monotone.

To facilitate a clear understanding of the nature of the fixed-point argument it is helpful to consider Proposition 17.C.2, in which boundary complications are eliminated by studying continuous, homogeneous of degree zero functions $z(p)$ satisfying Walras' law and defined for all nonnegative, nonzero price vectors. Within a framework of continuous and strictly convex preferences, this type of excess demand function is not compatible with monotone preferences but can arise with preferences that are locally nonsatiated. Recall also that the equilibrium condition when zero prices are allowed is $z(p) \leq 0$; see expression (17.B.1').

Proposition 17.C.2: Suppose that $z(p)$ is a function defined for all nonzero, nonnegative price vectors $p \in \mathbb{R}_+^L$ and satisfying conditions (i) to (iii) of Proposition 17.B.2 (i.e. continuity, homogeneity of degree zero and Walras' law). Then there is a price vector p^* such that $z(p^*) \leq 0$.

Proof: Because of homogeneity of degree zero we can restrict our search for an equilibrium to the unit simplex $\Delta = \{p \in \mathbb{R}_+^L : \sum_{\ell} p_{\ell} = 1\}$.

Define on Δ the function $z^+(\cdot)$ by $z^+(p) = \text{Max}\{z_{\ell}(p), 0\}$. Note that $z^+(\cdot)$ is continuous and that $z^+(p) \cdot z(p) = 0$ implies $z(p) \leq 0$.

Denote $\alpha(p) = \sum_{\ell} [p_{\ell} + z_{\ell}^+(p)]$. We have $\alpha(p) \geq 1$ for all p .

Define a continuous function $f(\cdot)$ from the closed, convex set Δ into itself by

$$f(p) = [1/\alpha(p)](p + z^+(p)).$$

Note that, corresponding to intuition, this fixed-point function tends to increase the price of commodities in excess demand. The construction of the function is illustrated in Figure 17.C.2 for the case $L = 2$.

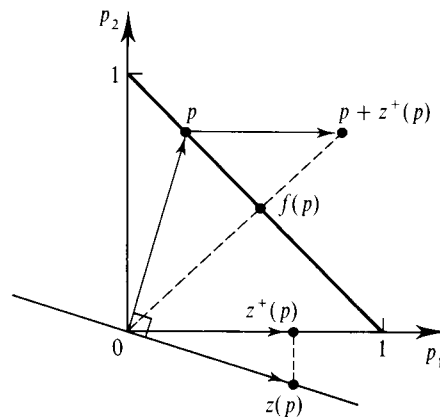


Figure 17.C.2
The fixed-point
function for
Proposition 17.C.2.

11. As we shall mention shortly in this section, existence would still obtain if $z(\cdot)$ was a convex-valued and upper hemicontinuous correspondence. In Figure 15.B.10(b), however, excess demand fails to satisfy the convex-valuedness property.

By Brouwer's fixed-point theorem (see Section M.I of the Mathematical Appendix) there is a $p^* \in \Delta$ such that $p^* = f(p^*)$. We show that $z(p^*) \leq 0$.

By Walras' law:

$$0 = p^* \cdot z(p^*) = f(p^*) \cdot z(p^*) = [1/\alpha(p^*)] z^+(p^*) \cdot z(p^*).$$

Therefore, $z^+(p^*) \cdot z(p^*) = 0$. But, as we have already pointed out, this implies $z(p^*) \leq 0$. ■

The applicability of Proposition 17.C.1 is not limited to exchange economies. We saw, for example, in Section 17.B (and Exercise 17.B.4), that if we allow for production sets that are closed, strictly convex, and bounded above (and if a positive aggregate consumption bundle is producible from the initial aggregate endowments), then the production inclusive excess demand function $\bar{z}(\cdot)$ satisfies conditions (i) to (v) of Proposition 17.B.2. Hence, Proposition 17.C.1 also implies that a Walrasian equilibrium necessarily exists in this case.

We also note for later reference that Proposition 17.C.1 holds as well for a convex-valued and upper hemicontinuous *correspondence* $z(p)$ that satisfies conditions (ii) to (v) (properly adapted) of Proposition 17.B.2. In this case, there exists a p such that $0 \in z(p)$. (See Exercise 17.C.2 for more on this point.)

Although Proposition 17.C.1 tells us that an equilibrium exists, it does not give us the equilibrium price vectors or allocations in an explicit manner. The issue of how to actually find equilibria was first considered by Scarf (1973). By now, a variety of useful techniques are available. They are very important for applied work, where the ability to compute solutions is key. See Shoven and Whaley (1992) for an account of applied general equilibrium.

The second welfare theorem of Section 16.D can be seen as a particular case of the current existence result. To see this, suppose that $x = (x_1, \dots, x_I)$ is a Pareto optimal allocation of a pure exchange economy satisfying the assumptions leading to Proposition 17.C.1. Then, by Proposition 17.C.1, a Walrasian equilibrium price vector p and allocation $\hat{x} = (\hat{x}_1, \dots, \hat{x}_I)$ exist for the economy in which endowments are $\omega_i = x_i$ for all i . Since x_i is affordable at prices p for every consumer i , we must have $\hat{x}_i \succeq_i x_i$ for all i . Hence, it follows from the Pareto optimality of $x = (x_1, \dots, x_I)$ that $\hat{x}_i \sim_i x_i$ for all i . But since $\hat{x}_i$ is consumer i 's optimal demand given prices p and wealth $w_i = p \cdot \omega_i = p \cdot x_i$, x_i must also be an optimal demand for consumer i for price-wealth pair $(p, p \cdot x_i)$. Hence, we see that the price vector p and the wealth levels $w_i = p \cdot x_i$ support the allocation x as a price equilibrium with transfers in the sense of Definition 16.B.4.¹²

17.D Local Uniqueness and the Index Theorem

Having established in Section 17.C (and Appendix B) conditions under which a Walrasian equilibrium is guaranteed to exist, we now begin a study of some issues related to its uniqueness or multiplicity.

12. The fact that the second welfare theorem can be viewed as a corollary of theorems asserting the existence of Walrasian equilibrium is valid much beyond the economies studied in this section.